

# From Ratings to Returns: Peer Information, Farmer Beliefs, and Technology Adoption under Performance Uncertainty

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## Abstract

Many smallholder farmers in low- and middle-income countries do not adopt modern agricultural inputs sold by agro-dealers due to uncertainty about their performance. We introduce a system that aggregates farmers' evaluations of seed into ratings and disseminates them to farmers and agro-dealers, providing a summary signal of seed performance based on peers' experiences. Such a system could affect outcomes through three channels: correcting farmers' beliefs and reducing their uncertainty about seed performance (belief updating), helping farmers identify better dealers (dealer sorting), or inducing dealers to improve seed preservation (supply-side response). We test this approach in a randomized controlled trial involving 3,500 farmers and 350 agro-dealers in the market for high-yielding maize seed in Uganda. We find that the intervention increases farmers' perceptions of seed performance, leads to higher adoption of high-yielding maize seed, and raises yields by 13%. We find no evidence that the intervention induces farmers to switch between dealers or improves dealer-level seed performance, providing little support for dealer sorting or supply-side quality improvement. The results point to belief updating as the primary mechanism. Consistent with this interpretation, effects on perceptions are larger among farmers without prior purchasing experience — those with the greatest uncertainty about seed performance.

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**Keywords:** agricultural technology adoption, agricultural input quality, performance uncertainty, beliefs, aggregated peer information, agro-dealers, small-holder farmers, improved seed, randomized controlled trial, Uganda

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# 1 Introduction

Smallholder farmers in low- and middle-income countries adopt productivity-enhancing agricultural inputs, such as improved seeds and inorganic fertilizers, at persistently low rates despite their potential to increase agricultural productivity, improve food security, and reduce poverty. As reviewed by [Suri and Udry \(2022\)](#), proposed explanations include credit, liquidity, and savings constraints; limited access to insurance; information frictions; high transaction costs and poor infrastructure; and imperfections in labor and land markets.

An important additional explanation is that farmers are uncertain about the performance of agricultural inputs supplied by agro-dealers, the primary distribution channel for improved seed varieties ([Michelson et al., 2021, 2023](#); [Dar et al., 2023](#)). For instance, in the market for high-yielding maize seed in Uganda, several studies document substantial performance uncertainty:<sup>1</sup> [Bold et al. \(2017\)](#) find that locally sold hybrid seed contains less than 50% authentic seed. [Barriga and Fiala \(2020\)](#) document improper handling and storage of maize seed purchased from retailers, while [Ilukor et al. \(2025\)](#) show that seed quality declines along the supply chain relative to samples obtained directly from seed companies. These studies document substantial uncertainty about the performance of seed sold through agro-dealers, but this performance is largely invisible to farmers at the point of purchase. This uncertainty directly discourages adoption by increasing the perceived riskiness of purchasing from agro-dealers. On the supply side, the same uncertainty may also weaken agro-dealers’ incentives: when buyers cannot credibly assess seed quality at the point of purchase, dealers face reduced reputational consequences for allowing seed to degrade through inadequate handling and storage.

Performance uncertainty also distorts the use of complementary inputs: [Bulte et al. \(2025\)](#) show that farmers who are unsure about input quality allocate less labor, reducing yields and making it harder to learn about the true profitability of improved varieties. More broadly, learning from own experience is difficult because agricultural outcomes depend on many factors beyond seed quality. [Laajaj and Macours \(2026\)](#) show that agricultural learning is inherently multidimensional and that even with intensive exposure, peer-to-peer transfer of knowledge about input performance faces severe frictions. These learning constraints allow poorly informed or biased beliefs to persist.

Credible signals that reduce performance uncertainty could address this, but existing mechanisms to signal seed performance at agro-dealers have major limitations. Seed companies signal quality through labeling and packaging, but farmers struggle to identify low-quality seed without specific training ([Gharib et al., 2021](#)). Regulatory certification systems are costly and, in Uganda, do not seem to provide a reliable signal of performance ([Tyack et al., 2026](#)). Moreover, these signals only reach farmers who already engage with agro-dealers, and even certified seed may perform poorly where

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<sup>1</sup>In this study, the term “high-yielding maize varieties” refers to both open-pollinated varieties (OPVs) and hybrids that have been genetically improved through breeding programs. We explain the terminology and our rationale for using this terminology in Appendix A.1.

handling and storage are inadequate (Barriga and Fiala, 2020).

In this paper, we test a decentralized alternative: we aggregate the experiences of farmers themselves into a signal of seed performance. Building on a well-established literature documenting the importance of peers in agricultural technology adoption (Foster and Rosenzweig, 1995; Bandiera and Rasul, 2006; Conley and Udry, 2010), we test whether aggregating peers’ experiences into a public signal of seed performance can improve market outcomes through three channels: belief updating about agro-dealer seed relative to outside options, dealer sorting conditional on purchasing from agro-dealers, or supply-side improvements in seed preservation. We introduce a system that collects farmers’ evaluations of maize seed purchased from agro-dealers and aggregates them into ratings that summarize the experiences of multiple farmers facing similar conditions. These ratings are disseminated to both farmers and agro-dealers, providing a summary signal about seed performance at agro-dealers that individual farmers may find difficult to obtain through their own experimentation or social networks.

Aggregated peer ratings address key limitations of existing approaches. They are based on farmers’ evaluations of experienced seed performance and therefore reflect performance under local conditions. They are disseminated directly to farmers, allowing the information to reach those who do not typically visit agro-dealers. The system is decentralized, relying less on institutional capacity. Importantly, the system does not measure or certify objective seed quality. It aggregates farmers’ assessments of experienced performance — whether the seed they purchased yielded well, germinated reliably, and tolerated local conditions. These assessments may diverge from laboratory measures of seed purity or authenticity.

A rating system that aggregates and disseminates farmers’ purchasing experiences could improve outcomes through several channels. First, farmers may hold imprecise or biased beliefs about seed performance. Disseminating aggregated peer experiences could reduce uncertainty among all farmers and, when priors are pessimistic relative to the signal, raise expected performance among farmers without prior purchasing experience, increasing adoption.

Second, dealer-specific ratings could enable dealer sorting by revealing performance differences across sellers conditional on purchasing from agro-dealers. In agricultural input markets, farmers typically choose among a small number of local dealers with limited information about relative seed performance. When dealer-choice frictions are high relative to quality variation, consumers settle for suboptimal matches (Stigler, 1961). By aggregating dispersed information about seller-specific performance, ratings enable the kind of sorting that characterizes online review platforms (Dranove and Jin, 2010; Luca, 2016). If performance varies substantially across dealers, ratings could induce farmers to switch to higher-performing sellers.

Third, the rating system could improve dealer-level seed performance by creating reputational incentives for agro-dealers to invest in seed preservation. In the prevailing market equilibrium, agro-dealers may have little incentive to invest in handling and storage if they believe buyers cannot credibly assess seed performance at the time of purchase and struggle to attribute outcomes such as yield to seed quality given the com-

plex and uncertain nature of agricultural production. By collecting and disseminating farmers’ experiences, the rating system makes dealers’ performance visible to potential buyers, strengthening the link between preserved seed performance and demand (Shapiro, 1982). These three channels are not mutually exclusive, but they generate distinct empirical predictions that allow us to assess their relative importance.

We test this approach in a randomized controlled trial (RCT) involving approximately 3,500 smallholder farmers and 350 agro-dealers in the market for high-yielding maize seed in eastern Uganda. The randomized design allows us to estimate causal effects, while three rounds of detailed data enable us to understand how and why the intervention affects behavior. We track outcomes along the full causal chain, from farmers’ perceptions and adoption decisions to agricultural productivity. Additional data on farmers’ choice of agro-dealers, agro-dealer seed handling practices, and objective indicators of seed performance allow us to trace the channels through which the intervention operates; measures of dealer services and signaling provide secondary evidence on other dealer responses.

Exposure to peer ratings improves farmers’ perceptions of seed performance and increases adoption of high-yielding maize seed. Treated farmers achieve 13% higher yields, showing that the intervention has economically meaningful downstream effects. Because adoption and productivity effects are consistent with all three channels, channel identification rests on the outcomes that differ across the model’s predictions. We find no evidence that the intervention induces farmers to switch between sellers, providing little support for dealer sorting as a primary channel. Nor do we find improvements in dealer-level seed performance, providing little support for supply-side quality improvement. Consistent with belief updating, effects on perceptions are larger among farmers without prior purchasing experience.

This paper contributes to the literature on agricultural technology adoption, particularly on the role of input quality uncertainty (Michelson et al., 2023) and farmers’ beliefs (Michelson et al., 2021; Hoel et al., 2024). We advance this literature in three ways. First, we introduce an intervention that addresses limitations of existing seed performance signals — it is decentralized, experience-based, and reaches farmers who do not engage with formal markets — and show that it improves adoption and productivity, with the pattern of results pointing to belief updating as the primary channel. Second, by tracking outcomes on both sides of the market — farmers’ beliefs and switching behavior, and agro-dealers’ seed handling practices and seed performance — we can assess the relative importance of belief updating, dealer sorting, and supply-side quality improvement. Together with Dar et al. (2023), our study is among the few to connect farmer and agro-dealer perspectives, providing a more complete picture than studies focusing on buyers or sellers alone. Third, the pattern of results — positive effects on perceptions and adoption, null effects on switching and dealer-level seed performance — provides evidence that uncertainty about seed performance is a binding constraint on adoption in seed markets. While overly pessimistic or imprecise beliefs have been documented in fertilizer markets (Michelson et al., 2021; Hoel et al., 2024), we provide evidence of this mechanism in seed markets.

More broadly, this paper connects to a growing literature on crowd-sourced quality information and market outcomes. Research on online review platforms documents that consumer feedback affects both sides of the market: buyers use reviews to make more informed choices, while sellers adjust their behavior in response to public quality signals (Bajari and Hortagsu, 2004; Jin and Kato, 2006; Dranove and Jin, 2010; Reimers and Waldfogel, 2021). In many of these settings, an important channel is sorting: reviews help consumers identify better sellers among many options. Our study shows that in a very different setting — a low-information, offline agricultural input market — crowd-sourced ratings need not operate primarily through seller choice. In our setting, the evidence points less to sorting across agro-dealers and more to belief updating about whether agro-dealer seed performs well relative to outside options. Hasanain, Khan, and Rezaee (2023) find a related null effect on switching in Pakistan, where crowd-sourced information on livestock insemination services does not induce farmers to switch providers.

The remainder of the paper is organized as follows. Section 2 presents the conceptual model, Section 3 describes the experimental design, Section 4 outlines the empirical strategy, Section 5 describes the data, Section 6 presents the results, and Section 7 concludes.

## 2 Conceptual model

This section develops a simple model of how aggregated peer information affects farmers and agro-dealer behavior and outcomes. The model generates a set of predictions that helps us organize the empirical analysis.

### 2.1 Setup

Let  $\theta_j$  denote the performance of seed sold by agro-dealer  $j$  relative to the farmer’s outside option. There are  $J$  agro-dealers in farmer  $i$ ’s area, each with performance  $\theta_j$ . This is the actual performance the farmer would realize if they planted the seed — it is unobservable to the farmer at the time of purchase. We decompose  $\theta_j$  as  $\theta_j = \theta_0 - d(e_j)$ , where  $\theta_0$  is the intrinsic performance of the seed upon arrival at agro-dealer  $j$  — determined upstream in the supply chain and outside the agro-dealer’s control and assumed to be the same for all agro-dealers<sup>2</sup> — and  $d(e_j) \geq 0$  captures performance degradation due to handling and storage, with  $d'(e_j) < 0$ . Higher agro-dealer effort  $e_j$  (e.g., proper storage containers, ventilation, temperature and moisture control) reduces degradation, preserving seed performance closer to its intrinsic level  $\theta_0$ .

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<sup>2</sup>This assumption abstracts from upstream variation in the seed received by agro-dealers. Allowing  $\theta_0$  to vary by dealer would introduce an additional source of dealer heterogeneity, but would not change the model’s distinction between belief updating, dealer sorting, and supply-side preservation effort.

Smallholder farmers choose between purchasing seed from agro-dealers and relying on outside options such as farmer-saved seed, informal exchanges, and seed distributed through public or non-governmental programs. Because seed performance is unobservable at the time of purchase, the decision depends on beliefs about the performance of agro-dealer seed relative to these alternatives. Seed has properties of a credence good: because yields depend on many factors beyond the seed itself, farmers cannot fully assess seed performance even after use (Laaaj and Macours, 2026). Individual learning about seed is therefore noisy, slow, and difficult to self-correct (Ahmad, 2022). This is compounded by the disproportionate salience of negative information (Rozin and Royzman, 2001; Vaish, Grossmann, and Woodward, 2008; Hornik et al., 2015), which may push beliefs below the true  $\theta$ . Evidence from fertilizer markets (Michelson et al., 2021; Hoel et al., 2024) and from maize seed in Uganda (Barriga and Fiala, 2020) is consistent with this hypothesis. That said, seed also has an experience good component (Foster and Rosenzweig, 1995): farmers who have purchased and planted agro-dealer seed can form more accurate beliefs than those who have not.

Farmers are risk-averse and face a two-part decision: whether to purchase seed from any agro-dealer rather than relying on the outside option (extensive margin), and if so, from which agro-dealer (agro-dealer choice). Farmer  $i$  evaluates each agro-dealer  $j$  by comparing the expected performance of that agro-dealer’s seed, discounted for uncertainty, against the cost of purchasing. Formally, the farmer purchases from the best available agro-dealer if the certainty equivalent exceeds the cost:

$$\max_j E_i[\theta_j] - \frac{\rho}{2}\sigma_{ij}^2 - c_j > 0 \quad (1)$$

where  $\sigma_{ij}^2$  is farmer  $i$ ’s subjective uncertainty about agro-dealer  $j$ ’s seed performance,  $\rho > 0$  is the coefficient of absolute risk aversion, and  $c_j > 0$  is the cost of purchasing from agro-dealer  $j$  relative to the outside option. Adoption depends on both the level of expected performance and subjective performance uncertainty: even at the same expected return, greater uncertainty discourages adoption.

## 2.2 Belief updating

The intervention can affect farmer outcomes through farmers’ beliefs about  $\theta_j$ . The farmer does not observe  $\theta_j$ ; instead, they hold a belief  $\mu_{ij} \equiv E_i[\theta_j]$  about each agro-dealer’s seed performance.

Because individual learning about seed performance is difficult (as discussed above), beliefs are likely to be imprecise or biased, and aggregated peer information can shape belief formation. The way it does so depends on farmers’ prior purchasing experience.

Farmer  $i$  holds a prior belief  $\mu_{ij}$  about each agro-dealer  $j$ :  $\theta_j \mid \mathcal{I}_i \sim \mathcal{N}(\mu_{ij}, \sigma_{ij}^2)$ . To generate predictions about heterogeneous treatment effects, we distinguish two types based on prior purchasing experience:

- **Purchasers** ( $i \in P$ ) have prior beliefs informed by own purchasing experience:  $\mu_{Pj}$  is close to  $\theta_j$ .

- **Non-purchasers** ( $i \in NP$ ) lack own purchasing experience: we assume  $\sigma_{NPj}^2 > \sigma_{Pj}^2$  (greater uncertainty), and allow  $\mu_{NPj}$  to differ from  $\theta_j$ . In the empirically relevant case where priors are pessimistic,  $\mu_{NPj} < \theta_j$ .

This captures the idea that, even though learning by doing is noisy, farmers with experience purchasing seed from agro-dealers have more precise priors.

The intervention provides farmer  $i$  with a signal about the performance of seed at agro-dealer  $j$ . The signal aggregates the experiences of  $n_j$  farmers who purchased from agro-dealer  $j$ :

$$s_j = \frac{1}{n_j} \sum_{k=1}^{n_j} (\theta_j + \varepsilon_k) = \bar{\theta}_j + \bar{\varepsilon}_j \quad (2)$$

where  $\varepsilon_k$  is the noise in rater  $k$ 's (drawn from the pool of previous purchasers) individual experience (reflecting weather, soil, management, and other factors beyond seed quality) and  $\bar{\varepsilon}_j \sim \mathcal{N}(0, \sigma_\varepsilon^2/n_j)$ .

Because raters are drawn from previous purchasers, the signal need not be representative of the experience that every farmer would have with agro-dealer  $j$ . Raters may differ from non-purchasers in land quality, management practices, complementary input use, or prior beliefs. We therefore interpret  $s_j$  as an informative peer-experience signal rather than an unbiased estimate of a universal agro-dealer-specific treatment effect. The model requires that, conditional on the local market and agro-dealer, average rater experiences contain information about agro-dealer seed performance beyond selection-driven differences across farmers.

Under this condition, averaging across raters reduces idiosyncratic noise: more raters produce a more precise signal about agro-dealer  $j$ 's seed performance. The Bayesian updating expression is a benchmark case in which farmers perceive the rating signal as credible and relevant; lower credibility or attention would attenuate the weight placed on  $s_j$  without changing the qualitative prediction that more uncertain priors update more strongly. Standard Normal-Normal bayesian updating yields posterior mean and variance:

$$\tilde{\mu}_{ij} = (1 - \alpha_{ij})\mu_{ij} + \alpha_{ij}s_j, \quad \tilde{\sigma}_{ij}^2 = \frac{\sigma_{ij}^2 \cdot \sigma_\varepsilon^2/n_j}{\sigma_{ij}^2 + \sigma_\varepsilon^2/n_j}, \quad \alpha_{ij} = \frac{\sigma_{ij}^2}{\sigma_{ij}^2 + \sigma_\varepsilon^2/n_j} \quad (3)$$

The weight  $\alpha_{ij}$  on the signal is increasing in prior subjective uncertainty. Since  $\sigma_{NPj}^2 > \sigma_{Pj}^2$ , non-purchasers place more weight on the signal and experience larger reductions in posterior uncertainty. The belief shift  $\tilde{\mu}_{ij} - \mu_{ij} = \alpha_{ij}(s_j - \mu_{ij})$  is upward when the signal is more favorable than the prior. For purchasers, the main effect is a precision gain (lower  $\tilde{\sigma}_{ij}^2$ ); for non-purchasers, the model predicts a larger precision gain and, when their priors are pessimistic relative to the signal, a larger upward revision in expected performance.

## 2.3 Dealer sorting

While the belief-updating channel captures changes in farmers' beliefs about agro-dealer seed relative to outside options, dealer sorting captures changes in the choice of seller conditional on purchasing from agro-dealers. Because the intervention provides agro-dealer-specific ratings ( $s_j$ ), it conveys two types of information. First, the average signal across agro-dealers,  $\bar{s} = \frac{1}{J} \sum_j s_j$ , provides information about the general performance of agro-dealer seed, allowing belief updating about agro-dealers overall. Second, the variation in signals across agro-dealers,  $s_j - \bar{s}$ , reveals performance differences and enables sorting.

Whether sorting occurs depends on the dispersion in  $\theta_j$  across agro-dealers. If performance varies substantially across agro-dealers and ratings reveal this variation, farmers can switch to higher-rated agro-dealers — predicting increased switching. If, however, ratings are similar across agro-dealers, the signal contains little agro-dealer-specific information, and the sorting channel is weak.

## 2.4 Supply-side responses

The intervention can also affect farmer outcomes by inducing agro-dealers to reduce degradation  $d(e_j)$ , raising  $\theta_j$  itself. This is because the rating system can create reputational incentives for agro-dealers. When ratings are disseminated to farmers, agro-dealers' performance becomes visible to potential buyers: their customers' experiences are collected, summarized, and shared.

Let  $e_j \geq 0$  denote agro-dealer  $j$ 's effort in seed preservation (e.g., proper storage containers, ventilation, temperature and moisture control), with convex cost  $c(e_j)$ . Recall that  $\theta_j = \theta_0 - d(e_j)$  with  $d'(e_j) < 0$ : higher effort reduces degradation, preserving seed performance closer to  $\theta_0$ . Without the intervention, agro-dealers' reputational incentives are limited to repeat purchases and informal word of mouth. Let  $R_C(\theta_j)$  denote the baseline demand for agro-dealer  $j$  that depends on quality through these informal channels, with  $R_C' > 0$ . Under the intervention, ratings are collected and disseminated, amplifying the reputational return to quality. Let  $R_T(\theta_j)$  denote demand under the rating system, with  $R_T'(\theta_j) > R_C'(\theta_j)$  — the marginal return to quality is higher when performance is publicly visible. The agro-dealer chooses effort to maximize:

$$\max_{e_j} R_\tau(\theta_0 - d(e_j)) - c(e_j), \quad \tau \in \{C, T\} \quad (4)$$

The first-order condition is  $R_\tau'(\cdot) \cdot (-d'(e_j^*)) = c'(e_j^*)$ . Since  $R_T' > R_C'$ , optimal effort is higher under the intervention:  $e_{jT}^* > e_{jC}^*$ . If agro-dealers respond to the intervention by investing in seed preservation, degradation falls and seed performance improves. If both channels operate, they may reinforce each other: belief updating increases farmers' willingness to engage with agro-dealers, while better seed preservation strengthens the actual purchasing experience.

## 2.5 Testable predictions

The model identifies three channels through which the intervention could affect outcomes: belief updating (Section 2.2), dealer sorting (Section 2.3), and supply-side quality improvement (Section 2.4). Each channel generates distinct predictions across five outcomes, summarized in Table 1.

- **Prediction 1 (Perceptions).** Under the belief updating channel, the treatment changes perceived seed performance in the direction of the signal, with stronger effects among non-purchasers. If prior beliefs are pessimistic relative to the signal, perceived seed performance shifts upward. Under the dealer sorting channel, there is no direct prediction on perceptions.<sup>3</sup> Under the supply-side channel, if dealers improve seed quality, perceptions may also improve, but the effect would not be concentrated among non-purchasers.
- **Prediction 2 (Adoption).** All three channels predict increased adoption: belief updating by reducing uncertainty and, when priors are pessimistic relative to the signal, by raising expected performance above the adoption threshold, dealer sorting by enabling farmers to identify better dealers, and supply-side responses by improving the objective quality of seed on offer. This prediction does not discriminate between channels.
- **Prediction 3 (Productivity).** All three channels predict higher yields, in part through increased adoption (Prediction 2). Dealer sorting and supply-side responses additionally raise yields conditional on adopting — through sorting to better agro-dealers and through improved seed quality, respectively. This prediction does not discriminate between channels. It requires that, on average, peer ratings move beliefs closer to the true performance of agro-dealer seed relative to outside options.
- **Prediction 4 (Switching).** The three channels differ in their predictions about switching between agro-dealers. Under belief updating, peer ratings provide information about the average performance of agro-dealer seed in the farmer’s area, shifting beliefs about agro-dealer seed in general rather than about any specific agro-dealer, predicting no increase in switching. Under the dealer sorting channel, dealer-specific ratings reveal performance differences, enabling farmers to switch to higher-rated dealers, predicting increased switching. The supply-side channel has no direct prediction on switching. Finding no increase in switching weakens evidence for dealer sorting as the primary channel.
- **Prediction 5 (Seed performance).** The three channels differ in their predictions about objective seed performance. Under belief updating, the intervention

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<sup>3</sup>If farmers sort to better dealers and subsequently experience higher seed performance, perceptions could improve indirectly over time. However, this would be a downstream consequence of switching behavior rather than an immediate effect of receiving ratings.

Table 1: Model predictions by channel and outcome

|                               | Belief updating                                     | Dealer sorting                      | Supply-side                         |
|-------------------------------|-----------------------------------------------------|-------------------------------------|-------------------------------------|
| Perceptions                   | Move toward signal, larger for non-purchasers       | No prediction                       | May improve, uniform across farmers |
| Adoption                      | Increases, concentrated among non-purchasers        | Increases, uniform                  | Increases, uniform                  |
| Productivity                  | Higher yields (via adoption + complementary inputs) | Higher yields (via better matching) | Higher yields (via better seed)     |
| Switching                     | No change                                           | Increases                           | No prediction                       |
| Dealer-level seed performance | No change                                           | No change                           | Improves ( $d(e_j)$ falls)          |

Note: Adoption effects alone do not discriminate between channels. Heterogeneity in perceptions (P1), limited switching (P4), and lack of dealer-level seed performance improvements (P5) jointly point to belief updating as the dominant channel.

shifts beliefs but not seed performance at a given agro-dealer:  $\theta_j$  is unchanged. Under dealer sorting, the intervention may change the performance of seed actually purchased by farmers if they reallocate toward higher-performing agro-dealers, but this operates through dealer choice rather than through product improvement at any given agro-dealer. Under the supply-side channel, if agro-dealers respond to reputational incentives by investing in handling and storage, degradation  $d(e_j)$  falls and dealer-level seed performance  $\theta_j$  improves. Finding no treatment effect on dealer-level seed performance is evidence against the supply-side channel.

The channels may also differ in their predicted timing of effects. Belief updating and dealer sorting operate immediately upon receiving ratings, predicting effects on perceptions and adoption from midline onward. Supply-side responses, by contrast, may take longer to translate into observable farmer outcomes, especially if quality improvements must pass through at least one rating cycle before becoming visible to farmers — a consequence of the credence good property of seed. This suggests that perceptions and adoption effects from supply-side responses would be more likely to appear at endline than at midline. We return to this distinction when interpreting the results.

## 3 Experimental design

### 3.1 Randomization and power

Randomization is conducted at the agro-dealer catchment-area level with equal assignment probability. Catchment areas consist of geographically proximate clusters of towns, villages, markets, trading centers, and other key hubs, and typically include several agro-dealers.<sup>4</sup> Randomizing at this level allows us to identify farmer-level treatment effects, as all farmers within a catchment area are exposed to the same information environment.

Sample sizes were determined using simulation-based power calculations based on data from prior surveys, allowing power to be assessed under empirically relevant outcome distributions at both the farmer and agro-dealer levels. Full details are provided in the [pre-analysis plan](#) (AEA RCT Registry, RCT ID 0006361).

### 3.2 Intervention

#### Rating collection and computation

Treated farmers were asked to rate maize seed sold by agro-dealers in their area. Farmers could rate an agro-dealer only if they had either personally purchased seed from that agro-dealer or knew someone who had, allowing them to report perceived seed performance based on direct or indirect experience. Enumerators guided farmers through the list of agro-dealers using a tablet-based application displaying names, location descriptions, and storefront pictures from the agro-dealer census (see Subsection 5.1).

Ratings were collected post-harvest, when farmers could form noisy assessments of seed performance based on realized outcomes such as germination, yield, drought resilience, pest and disease resistance, and crop maturation speed, and were elicited using the questions listed in Table 2 on a five-point star scale. The rating dimensions capture attributes that farmers attend to when purchasing and using seed, as identified through prior qualitative work, including workshops with experts from MAAIF, NARO, USTA, UNADA, Naseco Seed Company, and extension specialists.

Because agro-dealers stock heterogeneous seed varieties with different agronomic profiles, assessing overall seed performance in absolute terms may obscure trade-offs across characteristics. To address this, the rating instrument was designed to capture performance relative to product claims. Farmers were asked to rate attributes “as advertised,” e.g. by evaluating “maize seed sold by the agro-dealer” on “yield as advertised.” This approach allows performance to be assessed on a comparable basis across varieties

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<sup>4</sup>To define these areas, we used a haversine function to construct an adjacency matrix based on GPS coordinates, assigning agro-dealers less than five kilometers apart to the same catchment area. The five-kilometer threshold was selected based on a visual inspection of the map, the average village size in our sample, and the reported distance between farmers and agro-dealers in survey data from a previous study of the maize value chain, available [here](#).

Table 2: Questions for farmers to rate agro-dealers

|                                                                                                                                 | min    | max     |
|---------------------------------------------------------------------------------------------------------------------------------|--------|---------|
| Do you know this <i>shop name</i> or <i>dealer name</i> ,<br>sometimes called <i>nickname</i> , located in <i>market name</i> ? | no     | yes     |
| The place can be described as <i>description</i> .                                                                              |        |         |
| Please rate this agro-dealer on:                                                                                                |        |         |
| Quality and authenticity of seed                                                                                                | 1 star | 5 stars |
| Please rate the maize seed that this agro-dealer sells on:                                                                      |        |         |
| General quality                                                                                                                 | 1 star | 5 stars |
| Yield as advertised                                                                                                             | 1 star | 5 stars |
| Drought tolerance as advertised                                                                                                 | 1 star | 5 stars |
| Pest/disease tolerance as advertised                                                                                            | 1 star | 5 stars |
| Speed of maturing as advertised                                                                                                 | 1 star | 5 stars |
| Germination                                                                                                                     | 1 star | 5 stars |

while preserving an agro-dealer-level performance signal: an agro-dealer selling a variety without stress-tolerance traits that performs as advertised can receive a high rating, whereas an agro-dealer selling a stress-tolerant variety that underperforms relative to its claims receives a lower rating.

A potential concern is that the rating task itself could increase farmers’ awareness of agro-dealers, potentially confounding treatment effects. To address this, control-group farmers were also guided through the list of agro-dealers in their area, ensuring comparable exposure. However, they were not asked to provide ratings, as the act of rating is a component of the treatment. This design also reduces the scope for reactivity and experimenter bias.

Average ratings for each agro-dealer were computed by aggregating all farmer responses within the catchment area. Details on the rating computation are provided in Appendix A.2.

### Rating dissemination to farmers

The treatment relied on timely dissemination of agro-dealer ratings before the next agricultural season, allowing farmers to incorporate this information into seed purchasing decisions. Dissemination occurred through short message service (SMS) and in-person visits.

Farmers received one SMS per agro-dealer in their catchment area, delivered in one of three local languages – Lusoga, Lugwere, or Samia. For farmers in treatment areas, the message read:

Hello from AgroAdvisor! Did you know that customers rate the qual-

ity of maize seed sold at [*name of the agro-dealer*] as [*okay/good/very good/excellent*]?

Control farmers received a placebo message that merely highlighted the existence of agro-dealers in their catchment area.

In addition to SMS dissemination, enumerators revisited farmers and displayed agro-dealer ratings using a tablet-based application. The application cycled through all agro-dealers in the catchment area, presenting this message accompanied by a one-to-five-star visual rating:

We wanted to let you know that customers rate the quality of maize seed sold at [*name of the agro-dealer*] as [*okay/good/very good/excellent*]! The maize seed from this agro-dealer received a score of [*score*] out of 5!

For control group farmers, the application similarly cycled through agro-dealers in the catchment area without displaying ratings. Together with the placebo SMS, this design isolates the effects of rating information from potential reactivity or salience effects arising from reminding farmers of the existence of agro-dealers in their area.

### Rating dissemination to agro-dealers

Agro-dealers received their ratings in the form of a report. The front page displayed a certificate with a logo and the agro-dealer’s overall rating (Figure 1). The back page provided more detailed information, presenting ratings for seed sold by the agro-dealer across the six key dimensions and a table displaying average ratings of other agro-dealers in the same catchment area.

Agro-dealers were able to display the certificate. At endline, it was visible in 56% of shops. As a result, ratings may have been observed by non-sampled farmers, extending exposure beyond the study sample. Because randomization occurred at the agro-dealer catchment area level, any such additional exposure is contained within treated areas and does not threaten identification.

The study spans two agricultural seasons with three data collection rounds. At baseline (before season 1), we survey farmers and agro-dealers and collect ratings from treated farmers. Ratings are aggregated and disseminated to farmers and agro-dealers before planting in season 1. At midline (between seasons 1 and 2), we collect outcomes and a second round of ratings from treated farmers. Updated ratings are disseminated before planting in season 2. At endline (after season 2), we collect final outcomes. This design allows the belief updating channel to affect adoption from season 1 onward, while supply-side responses and any resulting changes in seed quality need at least one season to become observable to farmers through updated ratings.

## 4 Empirical strategy

We estimate intention-to-treat effects using Ordinary Least Squares:



Figure 1: SeedAdvisor certificate

$$Y_{ij} = \alpha + \beta T_j + \gamma' X_{ij} + \delta Y_{0ij} + \varepsilon_{ij} \quad (5)$$

where  $Y_{ij}$  is the outcome for farmer  $i$  in catchment area  $j$ ,  $Y_{0ij}$  is the corresponding mean-centered baseline outcome,  $T_j$  is a binary indicator for treatment assignment, and  $X_{ij}$  includes controls for orthogonal treatments in the factorial design, as the study was embedded in a larger project with additional cross-randomized interventions (demeaned and interacted with the main treatment effect; see [Lin, 2013](#); [Muralidharan, Romero, and Wüthrich, 2023](#)). The coefficient  $\beta$  captures the intention-to-treat effect.

We estimate analogous specifications for agro-dealer-level outcomes, where  $Y_{ij}$  denotes the outcome for agro-dealer  $i$  in catchment area  $j$ .

Since randomization occurs at the catchment area level, we cluster standard errors accordingly. For farmer-level outcomes, we use the conventional sandwich estimator without small-sample corrections. For agro-dealer-level outcomes, we apply the Bell-McCaffrey adjustment to account for the smaller number of observations ([Imbens and Kolesár, 2016](#)).

We trim continuous outcomes to limit the influence of outliers and apply inverse hyperbolic sine transformations to skewed variables. Outcomes with limited variation, defined as having 95% or more of observations at a single value within the relevant sample, are excluded from the analysis.

To account for multiple hypothesis testing, we aggregate related outcomes into summary indices following [Anderson \(2008\)](#), which assigns lower weights to highly correlated outcomes and thereby concentrates information. As a robustness check, we also com-

pute summary indices following [Kling, Liebman, and Katz \(2007\)](#). While these indices capture overall effects across outcome families, we also report treatment effects on individual outcomes to help interpretation, with individual significance treated cautiously.

Our primary identification strategy relies on the pattern of effects across outcomes: each channel generates a distinct empirical signature across perceptions, adoption, switching, and seed quality (Table 1). We additionally examine heterogeneity by prior purchasing experience. While this heterogeneity analysis is not pre-specified, it is motivated by the model’s prediction that farmers with greater uncertainty about seed performance should weight the peer signal more heavily. We present these results as exploratory.

## 5 Data

### 5.1 Sample

The agro-dealer sample was constructed by listing all agro-dealers across eleven districts in southeastern Uganda, drawing on official records of registered agro-dealers and extensive field visits to identify new and informal ones.

Following this census, the 348 eligible agro-dealers were assigned to 130 catchment areas (see Subsection 3.1 for details). Of the 130 catchment areas, 65 were assigned to treatment, covering 193 agro-dealers, and 65 to control, covering 155 agro-dealers. Across all catchment areas, the average number of agro-dealers per area was three, with the number ranging from one to 18.

To link agro-dealers with farmers, agro-dealers identified the villages where most of their customers reside. Enumerators then randomly sampled ten maize-growing households from these villages, yielding a sample of approximately 3,500 smallholder farmers.

Baseline data were collected from agro-dealers in September-October 2020 and from farmers in April 2021; midline data were collected in January-February 2022, and end-line data in July-August 2022.

At the agro-dealer level, enumerators interviewed the individual most knowledgeable about day-to-day operations, purchased a maize seed bag, and recorded indicators related to seed storage conditions. At the farmer level, enumerators interviewed the individual most knowledgeable about maize farming. In addition to general questions, farmers listed all maize plots, from which one was randomly selected for detailed data collection, yielding household-representative averages.

### 5.2 Baseline descriptives

This subsection describes the baseline sample. Table 3 reports farmer characteristics. The average household head is 49 years old, 78% are male, and 51% have completed primary education. Households have nine members on average and are located four kilometers from the nearest agro-dealer and nine kilometers from the nearest tarmac

road. Farmers cultivate three acres, have 23 years of maize-growing experience, and reported yields of 440 kilograms per acre in the season preceding the baseline survey.

Half of farmers planted high-yielding maize seed<sup>5</sup> on at least one plot in that season, with one in three purchasing it from an agro-dealer. The remainder primarily obtained high-yielding seed from Operation Wealth Creation/National Agricultural Advisory Services or non-governmental organizations, while some relied on their own retained seed or farmer-to-farmer exchanges. These patterns support Assumption 1, indicating that farmers have meaningful outside options to purchasing seed from agro-dealers.

68% of farmers report believing that maize seed sold by agro-dealers is adulterated. This widespread skepticism aligns with Assumption 3, which posits that farmers hold prior beliefs about seed performance at agro-dealers that leave room for belief updating to influence adoption decisions.

Table 4 reports baseline characteristics of agro-dealers and their shops. The average respondent is 32 years old, 60% are male, and over 90% have completed primary education. In 55% of cases, the respondent is also the owner of the agro-shop.

On average, agro-shops were established five years before the baseline survey, are located seven kilometers from the nearest tarmac road, and serve 41 customers per day. 74% sell only farm inputs and equipment, 60% provide credit, and 46% offer advisory services.

With the shop manager’s permission, enumerators inspected seed storage areas and recorded observable characteristics. They reported pest issues (e.g., rats or insects) at 65% of agro-shops and maize seed stored in open containers at 16% of agro-shops.

Enumerators also sampled maize seed bags at the agro-shops and measured their moisture content. Moisture is a commonly used objective indicator of seed performance because it directly affects germination and viability; additional details are provided in Appendix A.1. The average moisture content was 13.6%. Regarding packaging and labeling, 68% of purchased seed bags displayed a packaging date, 18% an expiry date, and only 8% a quality certification label issued by the National Seed Certification Services (NSCS),<sup>6</sup> consistent with weak baseline guarantees of seed performance provided through agro-shops (Assumption 2).

### 5.3 Randomization balance

We present standard orthogonality tables with pre-registered variables to assess baseline comparability between treatment and control groups. For farmer-level outcomes (Table 3), no comparison is significant; for agro-dealer-level outcomes (Table 4), two of 17 comparisons are significant at the 10% level. Joint F-tests further confirm baseline

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<sup>5</sup>“High-yielding maize varieties” refers to both open-pollinated varieties and hybrids that have been improved through breeding programs. “Farmer-saved seed” refers to seed of unknown progeny or quality, which typically performs worse than freshly purchased high-yielding maize varieties from agro-dealers. Our terminology and the rationale for these definitions are detailed in Appendix A.1.

<sup>6</sup>In Uganda, maize seed certification is overseen by the NSCS within MAAIF. Certified seed is identified by a blue tag or sticker, while quality-declared seed carries a green tag or sticker.

balance.

## 5.4 Attrition

Table 5 reports attrition rates for treatment and comparison groups. At midline, 12% of agro-dealers and 2% of farmers were missing; at endline, 14% of agro-dealers and 1% of farmers were not surveyed. To assess whether attrition was linked to treatment, we regressed the likelihood of leaving the sample on the treatment indicator and found that treated agro-dealers were significantly less likely to drop out.<sup>7</sup>

Differential attrition was further assessed by regressing pre-registered balance variables (see Table 4) on indicators for leaving the sample, for belonging to the control group, and on their interaction (Appendix A.3). Results indicate that attrition among agro-dealers is unlikely to bias our estimates. No differential attrition was observed at the farmer level; since primary outcomes are measured among farmers, attrition is unlikely to meaningfully affect the main treatment effects.

## 5.5 Rating data

At endline, the dyadic rating dataset comprises 16,823 farmer-dealer links. Among farmers who provided at least one rating, the average farmer rated 1.4 agro-dealers (min = 1, max = 10). The average agro-dealer was rated by 6.7 farmers (min = 0, max = 30).

Because ratings could be provided either by farmers who had personally purchased seed from the agro-dealer or by those who knew someone who had, the signal may reflect indirect information rather than firsthand performance experience. In practice, 93% of ratings were provided by farmers with direct purchase experience, and ratings are nearly identical between purchasers and non-purchasers (see Appendix A.4 for details). This suggests that the aggregate signal predominantly reflects direct experience.

Ratings may also reflect factors unrelated to seed performance, such as customer service or personal relationships. Appendix A.4 provides suggestive evidence that ratings are associated with observable characteristics plausibly related to seller and seed performance: specialized agro-shops and dealers with better seed handling practices receive higher ratings, and seed ratings on randomly selected fields are positively correlated with maize yields. These patterns are consistent with ratings reflecting farmers' assessments of seed performance rather than purely non-performance attributes.

Appendix A.4 provides additional detail on the rating data, including the distribution of ratings across the six performance dimensions and their inter-correlations.

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<sup>7</sup>Reasons for agro-dealer attrition at endline were: eleven shops closed, ten relocated, eight changed product offerings, three merged with other shops, one refused to be interviewed, and 17 reported other (4) or no specific reasons (13).

Table 3: Descriptive statistics and orthogonality tests - Farmer

|                                                                     | mean               | treated          | obs. |
|---------------------------------------------------------------------|--------------------|------------------|------|
| Household head's age in years <sup>†</sup>                          | 48.62<br>(13.38)   | -0.24<br>(0.56)  | 3453 |
| Household head is male <sup>†</sup>                                 | 0.78<br>(0.42)     | 0.03<br>(0.03)   | 3470 |
| Household head finished primary education <sup>†</sup>              | 0.51<br>(0.50)     | 0.04<br>(0.03)   | 3437 |
| Homestead's distance to nearest tarmac road in km                   | 9.39<br>(10.81)    | -1.23<br>(1.71)  | 3302 |
| Homestead's distance to nearest agro-dealer in km <sup>†</sup>      | 3.78<br>(4.79)     | 0.11<br>(0.37)   | 3339 |
| Number of people in household (incl. respondent) <sup>†</sup>       | 8.70<br>(3.98)     | -0.09<br>(0.18)  | 3470 |
| Number of rooms in house                                            | 3.49<br>(1.45)     | 0.02<br>(0.09)   | 3469 |
| Farmer's land for crop production in acres                          | 3.35<br>(4.32)     | 0.00<br>(0.22)   | 3442 |
| Years since farmer started growing maize                            | 23.09<br>(13.14)   | -0.55<br>(0.58)  | 3470 |
| Yield in kg/acre <sup>†</sup>                                       | 443.01<br>(304.99) | -6.14<br>(13.52) | 3308 |
| Farmer used high-yielding maize varieties on any plot <sup>†</sup>  | 0.49<br>(0.50)     | 0.01<br>(0.02)   | 3466 |
| Farmer bought this seed at agro-dealer <sup>†</sup>                 | 0.32<br>(0.47)     | 0.01<br>(0.02)   | 3424 |
| Amount of this seed farmer bought at agro-dealer in kg <sup>†</sup> | 9.52<br>(6.92)     | -0.34<br>(0.53)  | 1064 |
| Farmer thinks maize seed at agro-dealer is adulterated <sup>†</sup> | 0.68<br>(0.46)     | 0.00<br>(0.03)   | 2673 |
| Farmer used DAP/NPK                                                 | 0.25<br>(0.43)     | 0.02<br>(0.04)   | 3465 |
| Farmer used organic manure                                          | 0.07<br>(0.26)     | 0.01<br>(0.01)   | 3466 |

Note: Column (1) reports sample means at baseline and standard deviations below; column (2) reports differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; column (3) reports the number of observations; \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels. <sup>†</sup> indicates that the variable was pre-registered to test balance and is included in the F-test.

Table 4: Descriptive statistics and orthogonality tests - Agro-dealer

|                                                                   | mean                | treated                      | obs.   |
|-------------------------------------------------------------------|---------------------|------------------------------|--------|
| Respondent's age in years <sup>†</sup>                            | 32.43<br>(11.49)    | -2.24 <sup>+</sup><br>(1.21) | 347    |
| Respondent is male <sup>†</sup>                                   | 0.59<br>(0.49)      | -0.01<br>(0.06)              | 348    |
| Respondent finished primary education <sup>†</sup>                | 0.92<br>(0.27)      | -0.01<br>(0.03)              | 339    |
| Respondent owns shop                                              | 0.55<br>(0.50)      | 0.02<br>(0.06)               | 348    |
| Respondent received training on maize seed handling <sup>†</sup>  | 0.53<br>(0.50)      | 0.12 <sup>+</sup><br>(0.07)  | 348    |
| Respondent knows how to store seed after repackaging <sup>†</sup> | 0.27<br>(0.44)      | 0.08<br>(0.06)               | 348    |
| Agro-dealer's distance to nearest tarmac road in km <sup>†</sup>  | 6.56<br>(10.39)     | -1.58<br>(2.24)              | 343    |
| Agro-dealer only sells farm inputs                                | 0.74<br>(0.44)      | 0.03<br>(0.06)               | 348    |
| Years since agro-dealer establishment <sup>†</sup>                | 5.34<br>(6.30)      | 0.21<br>(0.78)               | 348    |
| Number of customers per day <sup>†</sup>                          | 41.49<br>(46.49)    | 6.43<br>(6.72)               | 346    |
| Quantity of maize seed sold in kg <sup>†</sup>                    | 695.50<br>(1497.18) | 176.31<br>(235.92)           | 340    |
| Amount of maize seed lost/wasted last season in kg <sup>†</sup>   | 3.50<br>(18.65)     | 2.40<br>(2.30)               | 348    |
| Number of maize varieties in stock                                | 2.83<br>(1.59)      | 0.19<br>(0.21)               | 344.00 |
| Agro-dealer has problem with pests                                | 0.65<br>(0.48)      | -0.03<br>(0.06)              | 347.00 |
| Agro-dealer stores maize seed in open containers                  | 0.16<br>(0.36)      | 0.08<br>(0.05)               | 347.00 |
| Agro-dealer received seed related complaint from customer         | 0.64<br>(0.48)      | 0.07<br>(0.05)               | 347.00 |
| Moisture in bag of maize seed in %                                | 13.56<br>(1.44)     | -0.18<br>(0.26)              | 347.00 |

Note: Column (1) reports sample means at baseline and standard deviations below; column (2) reports differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; column (3) reports the number of observations; \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels. <sup>†</sup> indicates that the variable was pre-registered to test balance and is included in the F-test.

Table 5: Attrition

|                             | <i>midline</i>   |                     | <i>endline</i>   |                    |
|-----------------------------|------------------|---------------------|------------------|--------------------|
|                             | mean             | treated             | mean             | treated            |
| Agro-dealer left the sample | 0.121<br>(0.326) | -0.108**<br>(0.035) | 0.144<br>(0.351) | -0.079+<br>(0.042) |
| Farmer left the sample      | 0.018<br>(0.134) | 0.001<br>(0.005)    | 0.008<br>(0.091) | -0.001<br>(0.003)  |

Note: Columns (1) and (3) report sample means at mid- and endline and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels.

## 6 Results

We present the treatment effects on outcomes at both the farmer and agro-dealer levels after one agricultural season (midline) and two seasons (endline). To address multiple hypothesis testing, outcomes are aggregated into indices following [Anderson \(2008\)](#), reported at the bottom of the tables.<sup>8</sup> As a robustness check, Appendix A.7 reports indices constructed following [Kling, Liebman, and Katz \(2007\)](#).

All outcome variables were pre-specified in a registered pre-analysis plan at the [AEA’s registry for RCTs](#). The full econometric analysis was run on simulated data prior to outcome data collection and documented in a [mock report](#). All documents, code, and data are publicly available in a version-controlled [GitHub repository](#).<sup>9</sup>

### 6.1 Perceived seed performance

Table 6 reports treatment effects on farmers’ perceived performance of maize seed sold by agro-dealers. Perceived performance is measured using an index that combines ratings across six performance dimensions for agro-dealers in the farmer’s catchment area, aggregated at the farmer level. Because the index requires farmers to provide non-missing ratings on all six dimensions for at least one agro-dealer, the estimation sample is smaller than the full survey sample. Treatment effects can be estimated only at endline, as only treated farmers provided ratings at baseline and midline.

The treatment increases the perception index by 0.09 standard deviations in the full sample, with the effect significant at the 10% level. We note that this test is con-

<sup>8</sup>In regressions using these overall indices, we do not control for baseline values, as doing so would restrict the sample to units with complete data for all constituent variables at baseline and the respective follow-up wave (midline or endline), thereby substantially reducing statistical power.

<sup>9</sup>The presentation of results differs slightly from the pre-registered structure due to a reordering of outcomes, which modifies the construction of some pre-specified indices but leaves the main conclusions unchanged. Analyses adhering strictly to the pre-analysis plan remain publicly available in the project repository (see, e.g., the archived version linked [here](#)).

Table 6: Treatment effects on farmers’ perceived seed performance

|                                                             | <i>baseline</i>  | <i>full sample</i>        |      | <i>sub-sample</i>         |      |
|-------------------------------------------------------------|------------------|---------------------------|------|---------------------------|------|
|                                                             | mean             | <i>endline</i><br>treated | obs. | <i>endline</i><br>treated | obs. |
| Index of farmer’s seed ratings of agro-dealers <sup>1</sup> | 0.000<br>(0.637) | 0.092+<br>(0.054)         | 1664 | 0.102+<br>(0.055)         | 996  |

Note: Column (1) reports the baseline mean and standard deviation below; column (2) reports the difference between treatment and control group and the standard error below; it is clustered at the level of randomization; column (3) reports the number of observations; columns (4) and (5) mirror this structure for the sub-sample of farmers who did not use high-yielding maize seed purchased from an agro-dealer on any plot at baseline, \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels; larger indices indicate more desirable outcomes.

<sup>1</sup>The index of farmer’s maize seed ratings contains 6 ratings: general quality, yield, drought tolerance, pest/disease tolerance, time of maturity, germination. The ratings are aggregated at farmer level (one farmer rates multiple agro-dealers), then the index is computed.

servative: the perception index requires non-missing ratings across all six dimensions, which substantially reduces the effective sample. A power calculation based on 123 catchment areas with an intra-cluster correlation of 0.165 yields a minimum detectable effect of approximately 0.14 ICW index units at 80% power ( $\alpha = 0.10$ ), compared to the observed effect of 0.09. The study is therefore underpowered for detecting effects of this magnitude on the perception outcome. A similar calculation for the non-purchaser sub-sample yields an MDE of 0.15 ICW index units; the larger point estimate in this subgroup (0.10) relative to the full sample (0.09) almost exactly compensates for the smaller sample, leaving the test equally powered. The fact that we nonetheless detect significant effects in both samples, with the subgroup pattern consistent with the theoretical prediction, is suggestive of a genuine perception shift, even if its precise magnitude is uncertain.

As an exploratory analysis, we examine heterogeneity by prior purchasing experience at agro-dealers, proxied by whether farmers used high-yielding maize seed purchased from an agro-dealer on any plot in the season preceding baseline. Consistent with the model’s prediction that farmers with greater uncertainty weight the signal more heavily, the treatment effect is larger among farmers without prior purchasing experience.

Overall, these findings are consistent with the rating information being more favorable than farmers’ prior beliefs, leading to an upward revision in the perceived performance of maize seed sold by agro-dealers.

## 6.2 Adoption and sourcing decisions

We examine whether the intervention affects seed adoption and sourcing decisions (Table 7). The treatment increases the [Anderson \(2008\)](#) index at both midline and endline,

with effects significant at the 5% level.

Focusing on sourcing decisions across all plots, treated farmers are about six percentage points more likely to report purchasing high-yielding maize seed from an agro-dealer at midline – an increase of 18% relative to the baseline mean. At endline, the effect declines to roughly three percentage points and is no longer statistically significant. We detect no effect on quantities purchased conditional on buying, indicating that the intervention operates primarily at the extensive rather than the intensive margin. Treated farmers are also more likely to report “using any high-yielding maize variety (such as an OPV or hybrid seed) on any of their plots,” with effects of 3.5 percentage points at midline and 4.2 percentage points at endline.

Evidence from a randomly selected plot confirms these patterns. The likelihood that seed was obtained from an agro-dealer increases by nearly five percentage points at midline – an increase of 14% relative to the baseline mean – and by four percentage points at endline. Treated farmers are correspondingly less likely to rely on farmer-saved seed, with a significant reduction at midline. Maize seed expenditure on the random plot also increases significantly. Effects on the use of hybrids or OPVs are positive but statistically insignificant.<sup>10</sup>

Taken together, the results show that exposure to aggregated peer ratings increases engagement with agro-dealers and shifts farmers away from farmer-saved seed.

Related to the effects on farmers’ adoption and sourcing decisions, we examine whether the treatment affected agro-dealers’ business activity (Table A.5). Sales volume is measured as the self-reported total quantity sold of the four most popular high-yielding maize varieties in the previous season (Longe 7H, Longe 10H, Longe 4, and Longe 5). Agro-dealers also reported the start-of-season sales price for each variety; we use the average of these prices. Revenue is calculated by multiplying quantities sold by prices and summing across varieties. Additional outcomes include the number of customers purchasing maize seed per day at the start of the season and the number of maize varieties stocked.

The intervention increases business activity. At midline, treated agro-dealers score significantly higher on the [Anderson \(2008\)](#) index, and revenue increases by nearly 20%, significant at the 10% level. The revenue increase coincides with a larger rise in sales volumes (28.4% relative to baseline) than in prices (2.3% relative to baseline), although neither component is individually statistically significant, suggesting that higher revenue primarily reflects increased demand rather than price adjustments. At endline, the overall index is significantly higher at the 1% level, driven primarily by a 31% increase in the number of customers served – approximately six additional customers per day.

The results show that increased farmer engagement translates into higher business activity among agro-dealers. Appendix A.5 reports business outcomes for two recently released maize varieties (Longe 10H and Longe 5), further supporting the positive impact on agro-dealer business activity.

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<sup>10</sup>Farmers reported the maize variety planted in the previous season. Varieties were classified as hybrid maize seed/OPV if they corresponded to improved types (e.g., Longe series, Bazooka, Panner, Wema, KH series) and the seed was newly purchased rather than recycled or farmer-saved.

Table 7: Treatment effects on farmers' adoption and sourcing decisions

|                                                                      | <i>baseline</i>          |  | <i>midline</i>     |      | <i>endline</i>    |      |
|----------------------------------------------------------------------|--------------------------|--|--------------------|------|-------------------|------|
|                                                                      | mean                     |  | treated            | obs. | treated           | obs. |
| Farmer bought seed at agro-input shop for any plot <sup>†</sup>      | 0.325<br>(0.468)         |  | 0.059**<br>(0.021) | 3145 | 0.031<br>(0.020)  | 3225 |
| Amount of this seed farmer bought at agro-input shop in kg           | 9.519<br>(6.920)         |  | -0.105<br>(0.358)  | 599  | 0.378<br>(0.431)  | 621  |
| Farmer used high-yielding seed on any plot <sup>†</sup>              | 0.492<br>(0.500)         |  | 0.035+<br>(0.020)  | 3206 | 0.042*<br>(0.020) | 3282 |
| Farmer bought seed at agro-input shop for specific plot <sup>†</sup> | 0.330<br>(0.470)         |  | 0.047*<br>(0.022)  | 3153 | 0.036+<br>(0.019) | 3240 |
| Farmer used farmer-saved seed on specific plot                       | 0.579<br>(0.494)         |  | -0.042+<br>(0.022) | 3153 | -0.016<br>(0.020) | 3240 |
| Cost of seed used on specific plot in UGX <sup>§†</sup>              | 14078.272<br>(24654.685) |  | 0.499*<br>(0.235)  | 2848 | 0.350+<br>(0.209) | 2942 |
| Farmer used hybrid seed/OPV on specific plot <sup>1†</sup>           | 0.432<br>(0.495)         |  | 0.035<br>(0.023)   | 2954 | 0.030<br>(0.023)  | 3047 |
| <b>Overall Anderson index</b>                                        | -0.013<br>(0.899)        |  | 0.087*<br>(0.042)  | 2854 | 0.086*<br>(0.039) | 2978 |
| Max. number of obs.                                                  |                          |  |                    | 3407 |                   | 3441 |

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels; <sup>†</sup> indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

<sup>1</sup>For this variable, only non-recycled (newly purchased, not farmer-saved) seed counted as hybrid variety/OPV.

<sup>§</sup>Due to the skewness of this variable, the regression was run after an inverse hyperbolic sine transformation. Coefficient estimates can therefore be interpreted as percentage changes. The baseline mean column shows the untransformed variable.

Table 8: Treatment effects on agro-dealers' business activity

|                                                      | <i>baseline</i>       |  | <i>midline</i>                |      | <i>endline</i>                 |      |
|------------------------------------------------------|-----------------------|--|-------------------------------|------|--------------------------------|------|
|                                                      | mean                  |  | treated                       | obs. | treated                        | obs. |
| Maize seed sales volume in kg <sup>§†</sup>          | 695.503<br>(1497.183) |  | 0.284<br>(0.227)              | 292  | 0.239<br>(0.253)               | 286  |
| Sales price of maize seed in UGX/kg <sup>†</sup>     | 4273.897<br>(955.073) |  | 99.272<br>(113.292)           | 275  | 145.861<br>(138.816)           | 264  |
| Revenue from maize seed in mln UGX <sup>§†</sup>     | 2.890<br>(6.286)      |  | 0.185 <sup>+</sup><br>(0.108) | 292  | 0.143<br>(0.118)               | 286  |
| Number of maize seed customers per day <sup>§†</sup> | 19.764<br>(20.689)    |  | 0.127<br>(0.101)              | 294  | 0.310 <sup>**</sup><br>(0.112) | 288  |
| Number of maize varieties in stock <sup>†</sup>      | 2.834<br>(1.589)      |  | 0.245<br>(0.245)              | 295  | 0.221<br>(0.220)               | 292  |
| <b>Overall Anderson index</b>                        | 0.031<br>(0.610)      |  | 0.197 <sup>*</sup><br>(0.092) | 274  | 0.238 <sup>**</sup><br>(0.082) | 270  |
| Max. number of obs.                                  |                       |  |                               | 306  |                                | 297  |

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels; <sup>†</sup> indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

<sup>§</sup>Due to the skewness of this variable, the regression was run after an inverse hyperbolic sine transformation. Coefficient estimates can therefore be interpreted as percentage changes. The baseline mean column shows the untransformed variable.

### 6.3 Productivity and harvest outcomes

We examine harvest outcomes for farmers (Table 9), measured on a randomly selected maize plot and including production, plot size, yield, the value of maize production at market prices net of seed costs, the quantity of maize retained for use as seed in the following season, and quantity sold, sales price, and total maize revenue.

The overall index shows a positive treatment effect at endline. Treated farmers achieve higher yields, increasing by 56 kilograms per acre (13% over the baseline mean, significant at the 1% level). The value of maize production net of seed costs is 66,000 UGX higher for treated farmers (approximately 18 USD at the time of the survey). Treated farmers retain less maize for seed at midline, indicating reduced reliance on farmer-saved seed.

We detect no effects on quantity sold or total maize revenue, consistent with the low baseline level of maize sales and suggesting that additional output is consumed within the household rather than sold.

A back-of-envelope calculation suggests that the adoption margin alone cannot fully account for these yield gains. Even assuming improved varieties yield 50 percent more than farmer-saved seed, the roughly 4 percentage point shift toward agro-dealer sourcing explains only about a quarter of the observed 56 kg/acre increase. However, the treatment also increases inorganic fertilizer use by approximately 5 percentage points (significant at both midline and endline) and reduces intercropping at midline. These complementary input responses are consistent with [Bulte et al. \(2025\)](#), who show that uncertainty about seed performance depresses investment in complementary inputs. By reducing such uncertainty, the ratings intervention encourages farmers to invest more in inputs whose returns depend on seed quality. The productivity gains therefore reflect both the direct effect of shifting toward improved varieties and increased complementary input use among farmers who are more confident about the seed they plant.

### 6.4 Dealer sorting and seed performance

The model predicts that dealer sorting should increase switching across agro-dealers, while supply-side quality improvement should raise dealer-level seed performance. We therefore examine these two diagnostic outcomes before turning to other dealer responses.

#### 6.4.1 Dealer sorting and agro-dealer switching

One alternative explanation for the treatment effects is that the intervention enables dealer sorting. The treatment may enable farmers to identify agro-dealers selling higher-performing products and switch to them, potentially leading to the exit of agro-dealers selling lower-performing products and thereby reducing adverse selection in the market. Similar dynamics are documented in settings where increased transparency allows buyers to screen sellers ([Jensen and Miller, 2018](#); [Lane, Schonholzer, and Kelley, 2022](#); [Cai,](#)

Table 9: Treatment effects on farmers' harvest on specific maize plot

|                                                   | <i>baseline</i>            |  | <i>midline</i>           |      | <i>endline</i>             |      |
|---------------------------------------------------|----------------------------|--|--------------------------|------|----------------------------|------|
|                                                   | mean                       |  | treated                  | obs. | treated                    | obs. |
| Production in kg <sup>†</sup>                     | 463.203<br>(399.595)       |  | -20.372<br>(14.529)      | 2884 | 43.937*<br>(17.922)        | 2898 |
| Area in acres                                     | 1.094<br>(0.655)           |  | -0.003<br>(0.029)        | 3004 | 0.006<br>(0.038)           | 3066 |
| Yield in kg/acre <sup>†</sup>                     | 443.013<br>(304.991)       |  | -23.006<br>(16.964)      | 2878 | 55.606**<br>(17.283)       | 2891 |
| Production minus cost of seed in UGX <sup>1</sup> | 307252.230<br>(294528.908) |  | -5960.653<br>(10229.650) | 2443 | 65621.966**<br>(22927.424) | 2525 |
| Amount kept as seed in kg <sup>§</sup>            | 14.506<br>(18.530)         |  | -0.188*<br>(0.092)       | 2931 | 0.036<br>(0.104)           | 2861 |
| Amount sold in kg <sup>§†</sup>                   | 195.295<br>(297.545)       |  | -0.201<br>(0.124)        | 3063 | 0.173<br>(0.173)           | 3137 |
| Sales price in UGX/kg                             | 506.954<br>(139.389)       |  | 33.027*<br>(14.244)      | 610  | 12.614<br>(41.238)         | 639  |
| Revenue in UGX <sup>§†</sup>                      | 97.783<br>(156.538)        |  | -0.393<br>(0.257)        | 3058 | 0.355<br>(0.363)           | 3109 |
| <b>Overall Anderson index</b>                     | -0.020<br>(0.784)          |  | -0.061<br>(0.039)        | 2932 | 0.097*<br>(0.041)          | 2900 |
| Max. number of obs.                               |                            |  |                          | 3407 |                            | 3441 |

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels; † indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

<sup>§</sup>Due to the skewness of this variable, the regression was run after an inverse hyperbolic sine transformation. Coefficient estimates can therefore be interpreted as percentage changes. The baseline mean column shows the untransformed variable.

<sup>1</sup>Production in UGX, measured as the number of maize bags harvested, multiplied by the market value per bag in UGX, minus the cost of maize seed used in UGX, calculated as the quantity of seed used in kg, multiplied by the cost per kg of seed in UGX.

Table 10: Treatment effects on farmers’ switching behavior

|                                   | <i>midline</i><br>mean | <i>midline</i><br>treated      | <i>midline</i><br>obs. | <i>endline</i><br>treated | <i>endline</i><br>obs. |
|-----------------------------------|------------------------|--------------------------------|------------------------|---------------------------|------------------------|
| Farmer switched to different shop | 0.590<br>(0.492)       | -0.062 <sup>+</sup><br>(0.033) | 1270                   | -0.031<br>(0.033)         | 1410                   |
| Max. number of obs. <sup>1</sup>  |                        |                                | 1270                   |                           | 1410                   |

Note: Column (1) reports midline means and standard deviations below (because this variable was not collected at baseline); columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels.

<sup>1</sup>Comparisons are restricted to farmers who purchased maize seed from an agro-input shop for any plot at mid- or endline and during the season before that (i.e., base- or midline).

Lin, and Szeidl, 2024; Hsu and Wambugu, 2024), as well as in a recent literature showing that reducing non-price search frictions can improve market functioning in low- and middle-income countries (Bai et al., 2020; Rudder and Dillon, 2024; Bergquist, McIntosh, and Startz, 2024; Startz, 2024; Dillon, Aker, and Blumenstock, 2025).

To assess whether the intervention operates through this alternative channel, we examine whether treated farmers are more likely to switch agro-dealers between seasons. The analysis is restricted to farmers who purchase maize seed from an agro-input shop for at least one plot in both the reference season (baseline or midline) and the subsequent season (midline or endline), as switching is only meaningful for farmers who purchase seed in both periods.

Results in Table 10 show that 59% of farmers who purchase maize seed at both baseline and midline switch agro-dealers between those two seasons. The treatment does not increase switching. If anything, treated farmers are slightly less likely to switch at midline, with the difference statistically significant at the 10% level. By endline, switching rates are statistically indistinguishable between treatment and control groups.

These results provide no evidence that the intervention reduces search frictions or induces reallocation across agro-dealers. Instead, the findings are more consistent with the belief-updating mechanism outlined in the model: the intervention appears to influence whether farmers engage with agro-dealers rather than which agro-dealer they choose. This finding is consistent with Hasanain, Khan, and Rezaee (2023), who implement a similar crowd-sourced information platform in Pakistan’s artificial insemination for livestock market and find no evidence that farmers switch to higher-performance veterinarians.

#### 6.4.2 Dealer-level seed performance

We also examine whether the intervention affects the underlying performance of seed sold by agro-dealers.

Table 11 reports treatment effects on seed performance. Enumerators randomly

purchased an original, sealed maize seed bag from agro-dealers and evaluated it based on moisture content, packaging integrity, and packaging information (e.g., visibility of the packaging date and lot number, and days since packaging). Consistent with the absence of changes in seed handling and storage practices, we find no significant treatment effects on seed performance indicators.

Appendix A.6 reports a power analysis for seed moisture content (a pre-registered objective measure) because not all sampled agro-dealers had maize seed in stock, showing that the observed effect size would not be detectable even with a much larger sample. As a second robustness check, we examine additional indicators of seed performance available for larger samples and again find no treatment effects.

Consistent with the absence of treatment effects on seed handling and storage practices reported below, the intervention did not improve the underlying performance of seed sold by agro-dealers. These findings provide little support for supply-side seed quality improvement as a primary channel.

## 6.5 Other dealer responses

Finally, we examine whether agro-dealers respond along dimensions other than dealer-level seed performance. These outcomes are useful for describing dealer behavior, but they are not the central test of the supply-side quality channel in the model. Evidence from other contexts shows that public quality disclosure can induce sellers to adjust behavior (Jin and Leslie, 2003; Klein, Lambertz, and Stahl, 2016; Annan, 2025).

Table 12 reports effects on agro-dealer practices. We construct two indices based on enumerator observations capturing labor-intensive practices (e.g., not storing seed in open containers) and capital-intensive practices (e.g., having a leak-proof roof). All variables included in the indices are listed in the table notes. Neither of the two indices responds to the intervention.

While the intervention does not affect seed handling and storage practices, there is suggestive evidence that agro-dealers respond along other dimensions. A third index captures service provision (e.g., recommending complementary inputs) based on agro-dealer-reported variables and is higher among treated agro-dealers, with the effect statistically significant at endline. A fourth index captures agro-dealer services reported by farmers, who likewise observe improved services among treated agro-dealers, with the effect statistically significant at midline.

We also find a positive effect on activities that signal credibility or professionalism to customers (e.g., membership in a professional association such as UNADA). A fifth index capturing signaling practices is higher among treated agro-dealers, with the effect statistically significant at endline. Because this result is less robust than the main demand-side findings, we treat it as suggestive rather than central to the interpretation.

Table 11: Treatment effects on agro-dealers' seed performance indicators

|                                                | <i>baseline</i>    |  | <i>midline</i>     |      | <i>endline</i>    |      |
|------------------------------------------------|--------------------|--|--------------------|------|-------------------|------|
|                                                | mean               |  | treated            | obs. | treated           | obs. |
| Moisture in % <sup>†</sup>                     | 13.564<br>(1.482)  |  | -0.122<br>(0.144)  | 175  | -0.220<br>(0.197) | 261  |
| Bag shows packaging date <sup>†</sup>          | 0.689<br>(0.464)   |  | 0.050<br>(0.072)   | 179  | 0.035<br>(0.064)  | 265  |
| Shelf-life in days <sup>1†</sup>               | 60.951<br>(40.960) |  | -8.272<br>(20.869) | 164  | 6.352<br>(8.289)  | 240  |
| Seed is in original undamaged bag <sup>†</sup> | 0.940<br>(0.238)   |  | 0.002<br>(0.046)   | 179  | 0.051<br>(0.055)  | 265  |
| Bag shows lot number <sup>†</sup>              | 0.508<br>(0.501)   |  | -0.001<br>(0.107)  | 179  | 0.027<br>(0.064)  | 265  |
| <b>Overall Anderson index</b>                  | 0.065<br>(0.364)   |  | 0.108<br>(0.103)   | 160  | 0.108<br>(0.090)  | 236  |
| Max. number of obs. <sup>2</sup>               |                    |  |                    | 179  |                   | 265  |

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels; <sup>†</sup> indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

<sup>1</sup>Days since the packaging date or, if the bag does not show the packaging date, days since the expiry date minus 6 months.

<sup>2</sup>The comparisons were only made for agro-dealers where the enumerator was able to buy a bag of maize seed at mid- or endline. Also, we do not control for the baseline values of the outcome variables in the entire table because only 144 of the 179 dealers who had seed at midline also had seed at baseline and only 183 of the 265 dealers who had seed at endline also had seed at baseline, so that controlling for baseline values would reduce the sample sizes drastically.

Table 12: Treatment effects on agro-dealers' practices

|                                                                  | <i>baseline</i>   |  | <i>midline</i>     |      | <i>endline</i>    |      |
|------------------------------------------------------------------|-------------------|--|--------------------|------|-------------------|------|
|                                                                  | mean              |  | treated            | obs. | treated           | obs. |
| Index of labor-intensive seed handling practices <sup>1†</sup>   | 0.010<br>(0.484)  |  | 0.099<br>(0.065)   | 285  | 0.074<br>(0.068)  | 274  |
| Index of capital-intensive seed handling practices <sup>2†</sup> | 0.000<br>(0.559)  |  | 0.000<br>(0.072)   | 270  | 0.057<br>(0.083)  | 265  |
| Index of services (self-reported by agro-dealers) <sup>3†</sup>  | 0.000<br>(0.382)  |  | 0.066<br>(0.060)   | 243  | 0.110*<br>(0.052) | 247  |
| Index of services (reported by farmers) <sup>4†</sup>            | -0.027<br>(0.583) |  | 0.301**<br>(0.069) | 259  | 0.086<br>(0.084)  | 271  |
| Index of quality signaling practices <sup>5†</sup>               | 0.002<br>(0.686)  |  | 0.153<br>(0.103)   | 267  | 0.265+<br>(0.147) | 255  |
| <b>Overall Anderson index</b>                                    | 0.034<br>(0.537)  |  | 0.375**<br>(0.113) | 162  | 0.178+<br>(0.106) | 177  |
| Max. number of obs. for dealer survey outcomes                   |                   |  |                    | 306  |                   | 297  |

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels; † indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

<sup>1</sup>The index of labor-intensive seed handling practices contains 6 variables: whether seed is stored in dedicated area, in correct lighting, on correct surface, not in open containers, whether shop has no pest problem, cleanliness and professionalism rating by enumerator.

<sup>2</sup>The index of capital-intensive seed handling practices contains 5 variables: whether roof is leak-proof, roof is insulated, walls are insulated, shop is ventilated, expired seed is handled correctly. All variables were observed or verified by enumerators, except the last, which is self-reported.

<sup>3</sup>The index of services (self-reported) contains 8 variables: whether shop offers explanations, complementary input recommendations, extension/training, discounts for larger quantities, credit, did not receive seed related customer complaint, accepts mobile money, stocks small (not repackaged) seed package.

<sup>4</sup>The index of services (reported by farmers) contains 7 variables: whether shop offers refund/insurance, credit, training/advice, delivery, after-sales service, accepts different payment methods, sells small quantities. Farmers' responses are aggregated at dealer level, then the index is computed.

<sup>5</sup>The index of quality signaling practices includes 5 variables: whether shop displays official certificate, is registered with UNADA, is a member of another professional association, holds a trading license issued by local government, number of shop inspections.

## 7 Conclusion

This study examines how aggregated peer ratings of seed performance affect farmer beliefs, agro-dealer choice, and agro-dealer behavior. The intervention could operate through three channels: correcting pessimistic or imprecise beliefs among farmers (belief updating), helping farmers identify and switch to better-performing dealers (dealer sorting), or inducing agro-dealers to invest in seed preservation (supply-side quality improvement). Using a randomized experiment, we evaluate the effects of this intervention on a range of farmer and agro-dealer outcomes and distinguish between these channels.

We find that the intervention leads farmers to update their beliefs and improves their perceptions of seed performance. In an exploratory analysis, we find that effects on perceptions are larger among farmers without prior purchasing experience, consistent with the model’s prediction that farmers facing greater uncertainty about seed performance weight the peer signal more heavily.

As farmers update their beliefs, the intervention increases their adoption of high-yielding maize varieties purchased from agro-dealers and leads to higher productivity and improved harvest outcomes. The combination of positive treatment effects on perceptions, adoption, and productivity suggests that prior beliefs were overly pessimistic relative to seed performance and that the seeds accessed through agro-dealers performed favorably compared with outside options. Consistent with the model, these findings indicate that aggregated peer ratings can move farmers’ beliefs closer to the performance of seed sold through agro-dealers, shifting adoption decisions toward the welfare-optimal level.

This finding is consistent with existing evidence. Studies from fertilizer markets show that farmers often underestimate input quality relative to objective measures (Michelson et al., 2021; Hoel et al., 2024), and recent work on maize seed in Uganda suggests that farmers’ perceptions may be more pessimistic than warranted by measured performance (Barriga and Fiala, 2020). Insights from psychology and economics further suggest that negative information tends to be more salient, spreads faster, and receives disproportionate weight in social learning under noisy production conditions, which can skew belief formation toward pessimism (Rozin and Royzman, 2001; Vaish, Grossmann, and Woodward, 2008; Conley and Udry, 2010; Hornik et al., 2015).

On the supply side, the evidence is more limited. We find no improvement in seed handling, storage practices, or objective seed quality. There is suggestive evidence of some changes in service provision and signaling professionalism, but these effects are less robust than the demand-side findings and are not central to our interpretation. We also find no evidence that the intervention induces farmers to switch between agro-dealers, providing little support for dealer sorting or supply-side quality improvement as primary channels. The pattern of results — positive effects on perceptions and adoption, null effects on switching and seed quality — points to belief updating as the primary impact pathway.

This study highlights the potential for aggregated peer information to improve mar-

ket outcomes in environments where buyers face uncertainty about input performance and learning from own experience and social networks is limited. By pooling experiences from multiple users, such signals can help farmers form more accurate beliefs about input performance.

More broadly, our findings suggest that the primary channel through which crowd-sourced information platforms affect market outcomes depends on the nature of the binding constraint. In markets where consumers already participate but face difficulty choosing among sellers, the main channel is sorting — as documented in the online reviews literature. In markets where the binding constraint is participation itself — as in our setting, where many farmers do not engage with agro-dealers due to uncertainty about seed performance — the main channel is belief updating at the extensive margin. This distinction has implications for how clearinghouse systems should be designed: when the goal is to increase market participation rather than improve matching, broad dissemination to non-participants matters more than fine-grained seller rankings.

These findings have implications for policy and practice. Aggregating and disseminating users' experiences can improve market outcomes when quality assurance systems are weak or imperfect, offering a decentralized complement that does not rely on strong institutional capacity. Peer ratings help farmers form more accurate beliefs, reducing the barrier to adoption. There is also suggestive evidence that the accountability pressure from ratings may incentivize service improvements among agro-dealers, though this channel requires further investigation. For private actors, including agro-dealers and seed companies, experience-based feedback mechanisms can build credibility and strengthen market engagement.

At the time of implementation, limited smartphone access required an offline design. As digital access expands among smallholder farmers, similar systems could be implemented through online platforms, offering lower costs, broader reach, and greater scalability, although this remains to be tested.

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# A Appendix

## A.1 Context on seeds and their quality

In this study, the term “high-yielding maize varieties” refers to both open-pollinated varieties (OPVs) and hybrids that have been genetically improved through public or private sector breeding programs. While the difference between these two types of maize is based on their reproductive biology, there are also economic implications ([Morris, Rusike, and Smale, 1998](#); [Spielman and Kennedy, 2016](#)). We explain the terminology and our rationale for using this terminology below, recognizing that our explanation is a simplification of much more complex biological phenomena.

Maize hybrids tend to confer a significant advantage in terms of vigor – robust growth, high yields, and uniform yield performance, even under suboptimal conditions. This is due to the expression of heterosis that results from crossing distinct inbred parent lines to produce genetically superior progeny (F1) that is planted by the farmer. However, this vigor declines dramatically if farmers save and plant grain that was set aside from harvest. Instead, the farmer must purchase fresh (F1) seed each season to continue to realize the benefits of heterosis. The combination of vigor and purchase frequency can incentivize public investment in hybrid maize improvement, seed production, and marketing.

OPVs, on the other hand, are typically less vigorous than hybrids. Although they are also less susceptible to genetic degeneration and can be saved and planted across several generations without loss of vigor, their yield performance is typically lower than hybrids. Moreover, OPV seed saving practices – and thus less frequent purchases of fresh seed – tend to reduce these private investment incentives.

Relatedly, we use the term “farmer-saved seed” in this study to refer to seed that is of an unknown progeny or quality. They may be seed saved from the harvest of F1 hybrids or OPVs, and they may be  $n$  generations old. They may have been stored under suboptimal conditions between seasons, exposed to moisture, pests, or diseases, and, as a result of both genetic and physical factors, perform poorly relative to fresh hybrid or OPV seed purchased from an agro-input shop.

As highlighted in a recent review by [Michelson et al. \(2023\)](#), seed quality is complex and multi-dimensional due to the living nature of seeds, encompassing three essential dimensions. First, analytical purity measures the proportion of a seed sample that belongs to the intended species. Second, varietal purity ensures the seed is of the expected variety, possessing the claimed genetic characteristics. Third, viability and performance assess the seed’s capacity to germinate and grow with sufficient vigor.

This study focuses on the viability and performance dimension of maize seed quality. From a farmer’s perspective, this dimension is particularly salient, as it represents a crucial outcome that farmers observe during germination, early crop growth, and harvest, although it can be difficult to attribute realized outcomes to seed performance alone. Moreover, while some storage and handling practices, such as seed mixing or mislabeling, may affect varietal purity, the practices examined in this study primarily

influence seed viability and performance.

Objectively assessing maize seed performance is complex, but measuring the moisture content of a seed sample is the most common and practical test. Moisture is a powerful predictor, as it directly affects germination outcomes. [Afzal et al. \(2017\)](#), e.g., report germination rates of approximately 85% for seeds with a moisture content of 12.7%, compared to just 50% for the same seeds at 15.6%. Prolonged exposure to high humidity rapidly degrades seed viability by increasing susceptibility to damage from pathogens, insects, and physical handling ([Michelson et al., 2023](#)). The moisture content of a seed sample reflects recent storage conditions – whether the seeds were kept in airtight packages or exposed to open air. Consulted experts emphasized a widely used rule of thumb: each one-percentage-point increase in moisture content halves the seed’s shelf life.

While moisture measurement is standard practice among seed companies and crop inspectors and provides a valuable indication of seed performance, it is inherently one-dimensional. Additional quality assessments include germination tests conducted in controlled chambers; vigor tests under the growing conditions that may occur in the field; grow-out tests, which evaluate genetic purity through morphological analysis; seed health testing which measures the presence of pathogens and diseases, and genotyping (DNA fingerprinting), which verifies genetic purity using single nucleotide polymorphism analysis.

## A.2 Rating computation

**What if a treated agro-dealer does not receive a single rating?** If a treated agro-dealer is not rated by any farmer – e.g., because no sampled farmer had personally purchased seed there or knew someone who had – we informed farmers that we had no information about the shop while still mentioning its existence. In the first round, 16 out of 193 treated agro-dealers were not rated by any farmer.

**Should more ratings lead to better ratings?** Some agro-dealers were not rated by any farmer in the first round, while others received up to 22 ratings. If agro-dealer A is rated by ten farmers with an average of 3.5 and agro-dealer B is rated by one farmer with a rating of 3.6, agro-dealer B is treated as having the higher rating.

**Should ratings reflect catchment area performance?** Ratings were highly concentrated at the upper end of the scale, with fewer than 9% of agro-dealers receiving a rating below three out of five. Ratings are therefore transformed into quintiles based on their distribution. To avoid conveying an overly negative signal, the quintile labels were chosen to carry positive connotations.

**Are female agro-dealers rated worse than male agro-dealers?** Significant differences were observed between the ratings of female agro-dealers (41% of the sample) and male agro-dealers (59%), even after controlling for potential confounders such as education and multiple performance indicators. These differences are unlikely to reflect true disparities in performance and instead appear to stem from biased perceptions, likely driven by discrimination (De, Mieke, and Van Campenhout, 2024). To mitigate potential harm from the intervention, the ratings of female agro-dealers were adjusted accordingly. Specifically, each seed performance attribute was regressed on a gender dummy, and the resulting coefficients were added to the initial ratings of female agro-dealers.

### A.3 Differential attrition tests

To examine differential attrition, we regress the pre-registered balance variables (see Table 4) on a binary variable indicating whether an agro-dealer left the sample, a binary variable indicating whether an agro-dealer belonged to the control group, and their interaction. Column (2) reveals that agro-dealers who left the sample at midline operated closer to the nearest tarmac road – the only significant difference between those who left and those who stayed when attrition was most pronounced (see Table 5). By endline, however, those who left were younger, more likely to be female, had less established businesses, were more likely to operate specialized shops, and still tended to be closer to a tarmac road, see column (5). This suggests that agro-dealers who left differed from those who remained, regardless of exposure to the treatment, possibly reflecting greater vulnerability during the COVID-19 period, leading to business closures. Columns (3) and (6) show minimal differences between control and treated agro-dealers, consistent with Subsection 5.3. Examining the interaction between the indicators reveals few differences between agro-dealers who left the sample *and* belonged to the control group and those who did not, suggesting that differential attrition is unlikely to bias our estimates.

Table A1: Differential attrition tests - Agro-dealer

|                                                 | <i>baseline</i>     |                    | <i>midline</i>               |                      | <i>endline</i>               |                             |
|-------------------------------------------------|---------------------|--------------------|------------------------------|----------------------|------------------------------|-----------------------------|
|                                                 | mean                | attrited           | belongs to control           | attrited and control | belongs to control           | attrited and control        |
| Respondent's age in years                       | 32.43<br>(11.49)    | -2.50<br>(2.25)    | 2.68 <sup>+</sup><br>(1.40)  | -1.29<br>(3.04)      | -5.59**<br>(1.27)            | 5.45 <sup>+</sup><br>(3.04) |
| Resp. is male                                   | 0.59<br>(0.14)      | -0.17<br>(0.06)    | 0.05<br>(0.17)               | -0.12<br>(0.49)      | -0.34**<br>(0.10)            | 0.28 <sup>+</sup><br>(0.14) |
| Resp. finished primary education                | 0.92<br>(0.27)      | 0.01<br>(0.08)     | 0.01<br>(0.04)               | -0.02<br>(0.10)      | 0.04<br>(0.05)               | -0.04<br>(0.08)             |
| Resp. owns shop                                 | 0.55<br>(0.50)      | -0.15<br>(0.17)    | -0.01<br>(0.06)              | 0.01<br>(0.20)       | -0.05<br>(0.16)              | 0.14<br>(0.20)              |
| Resp. received training on maize seed handling  | 0.53<br>(0.50)      | 0.14<br>(0.11)     | -0.11<br>(0.07)              | -0.18<br>(0.16)      | 0.10<br>(0.10)               | -0.19<br>(0.15)             |
| Resp. knows how to store seed after repackaging | 0.27<br>(0.44)      | 0.13<br>(0.17)     | -0.08<br>(0.06)              | -0.10<br>(0.19)      | 0.03<br>(0.12)               | 0.03<br>(0.15)              |
| Dealer's distance to nearest tarmac road in km  | 6.56<br>(10.39)     | -5.37**<br>(1.37)  | 1.68<br>(2.33)               | 2.63<br>(2.63)       | -3.38 <sup>+</sup><br>(1.74) | 0.92<br>(2.93)              |
| Dealer only sells farm inputs                   | 0.74<br>(0.44)      | 0.11<br>(0.10)     | 0.00<br>(0.07)               | -0.18<br>(0.14)      | 0.23**<br>(0.07)             | -0.27*<br>(0.12)            |
| Years since dealer establishment                | 5.34<br>(6.30)      | -1.93<br>(1.26)    | 0.07<br>(0.81)               | -0.35<br>(1.78)      | -2.99*<br>(0.96)             | 0.60<br>(1.51)              |
| Number of customers per day                     | 41.49<br>(46.49)    | -7.99<br>(8.66)    | -5.02<br>(7.50)              | -5.48<br>(10.59)     | -8.05<br>(7.01)              | -4.79<br>(9.53)             |
| Quantity of maize seed sold in kg               | 695.50<br>(1497.18) | 234.18<br>(833.31) | -144.71<br>(376.10)          | -847.11<br>(875.12)  | 125.10<br>(459.81)           | -551.67<br>(483.08)         |
| Maize seed lost/wasted last season in kg        | 3.50<br>(18.65)     | 2.73<br>(8.06)     | -2.22<br>(2.35)              | -3.06<br>(8.16)      | 3.91<br>(9.08)               | -5.89<br>(9.21)             |
| Dealer has problem with pests                   | 0.65<br>(0.48)      | 0.16<br>(0.11)     | 0.08<br>(0.06)               | -0.36*<br>(0.17)     | -0.02<br>(0.09)              | -0.08<br>(0.14)             |
| Dealer stores maize seed in open containers     | 0.16<br>(0.36)      | -0.05<br>(0.10)    | -0.09 <sup>+</sup><br>(0.05) | 0.09<br>(0.12)       | -0.05<br>(0.10)              | 0.09<br>(0.12)              |
| Dealer received seed related customer complaint | 0.64<br>(0.48)      | -0.03<br>(0.12)    | -0.04<br>(0.05)              | -0.14<br>(0.17)      | -0.11<br>(0.12)              | 0.05<br>(0.16)              |
| Moisture in bag of maize seed in %              | 13.56<br>(1.44)     | -0.42<br>(0.52)    | 0.10<br>(0.27)               | 0.27<br>(0.66)       | -0.45<br>(0.31)              | 0.48<br>(0.56)              |

Note: Column (1) reports sample means at baseline and standard deviations below; columns (2) and (5) report differences between agro-dealers who attrited and those who did not, and standard errors below; columns (3) and (6) report differences between control and treatment agro-dealers, and standard errors below; columns (4) and (7) report differences between agro-dealers who attrited and belong to the control group and those who do not, and standard errors below; all standard errors are clustered at the level of randomization; \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels.

## A.4 Agro-dealer rating patterns

**Comparison of rating dimensions** We asked farmers to evaluate maize seed sold by an agro-dealer across six dimensions: General quality, Yield as advertised, Drought tolerance as advertised, Pest/disease tolerance as advertised, Speed of maturing as advertised, and Germination. Figure A1 presents the distribution of farmers’ ratings for these attributes along with their mean values.

The figure reveals variation across attributes, with some showing more concentrated distributions and others exhibiting greater dispersion. The attributes most strongly correlated with the *General quality* rating are *Yield as advertised* (0.54) and *Germination* (0.38). These two attributes also have the smallest mean absolute differences (0.47 for *Yield as advertised* and 0.55 for *Germination*), suggesting that these are the attributes farmers primarily consider when assessing overall performance.

**Comparison of ratings by purchase channel** Farmers could rate an agro-dealer if they met one of two conditions: either they had purchased seed from the agro-dealer themselves or they knew someone who had. Ideally, only farmers with firsthand experience would provide ratings, but restricting responses in this way risked generating too few observations for a reliable signal. While this decision was made at the experiment’s design stage and cannot be revised ex post, we can compare endline ratings between farmers who purchased seed from the agro-dealer and those who did not but knew someone who had.

Fortunately, the vast majority (93%) of ratings came from farmers who had purchased seed, while only 7% were provided by those without direct purchase experience. This alleviates concerns that ratings are primarily driven by reputation or hearsay effects. While the mean ratings differ slightly (3.6 for farmers who bought seed vs. 3.7 for those who did not), the density plots are similar (see Figure A2).

**Correlating ratings and performance indicators** There is concern that small-holder farmers may base their ratings of agro-input shops on factors such as customer service rather than seed performance or rely on the agro-dealer’s reputation or personal relationship with them. To address this, we test whether the ratings – specifically, the index of endline ratings computed following [Anderson \(2008\)](#) – correlate with objective indicators of seller and seed performance, also measured at endline.

We find that specialized agro-input shops, which exclusively sell farm inputs, receive higher ratings, as do dealers with better seed handling practices, objectively observed by our enumerators (Table A2). Also, farmers not only rated seeds purchased from agro-input shops but also those used on randomly selected maize fields, using the same questions. These seed ratings are positively correlated with crop yields (Table A3).

While alternative explanations for these significant correlations exist, the overall pattern supports our claim that farmers are assessing seed performance. The ratings therefore appear to provide an informative signal, although farmers’ perceptions may not be perfectly accurate given the many factors influencing agricultural production.

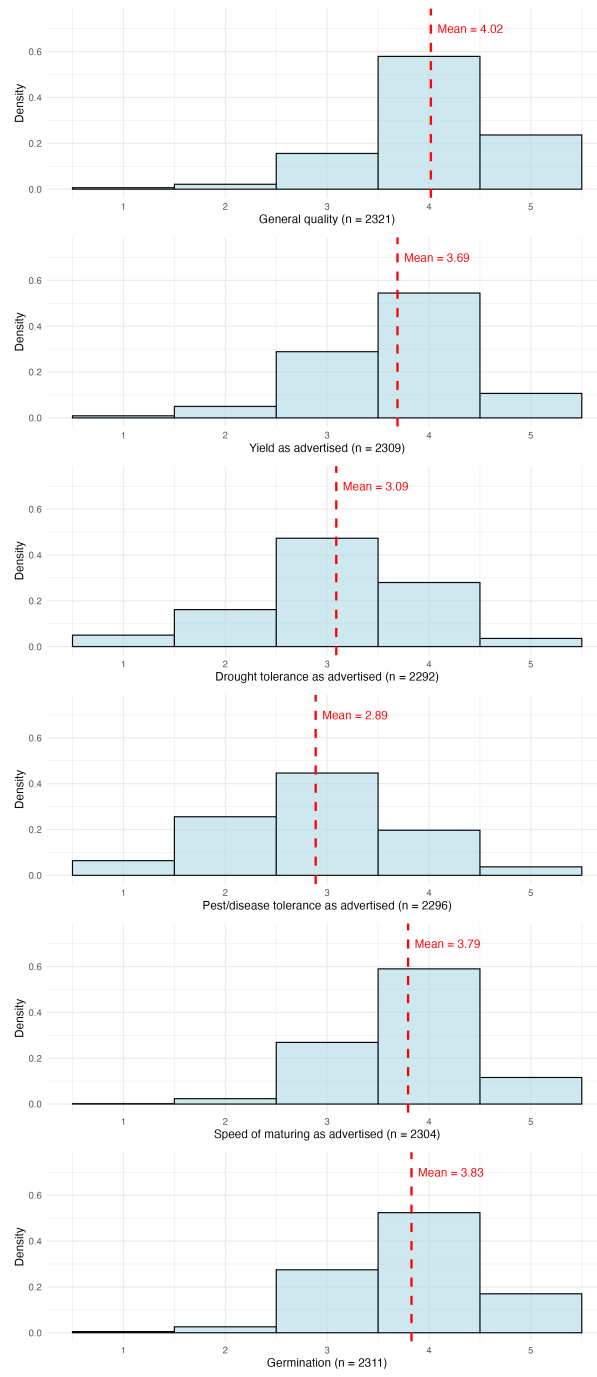


Figure A1: Comparison of rating dimensions at endline

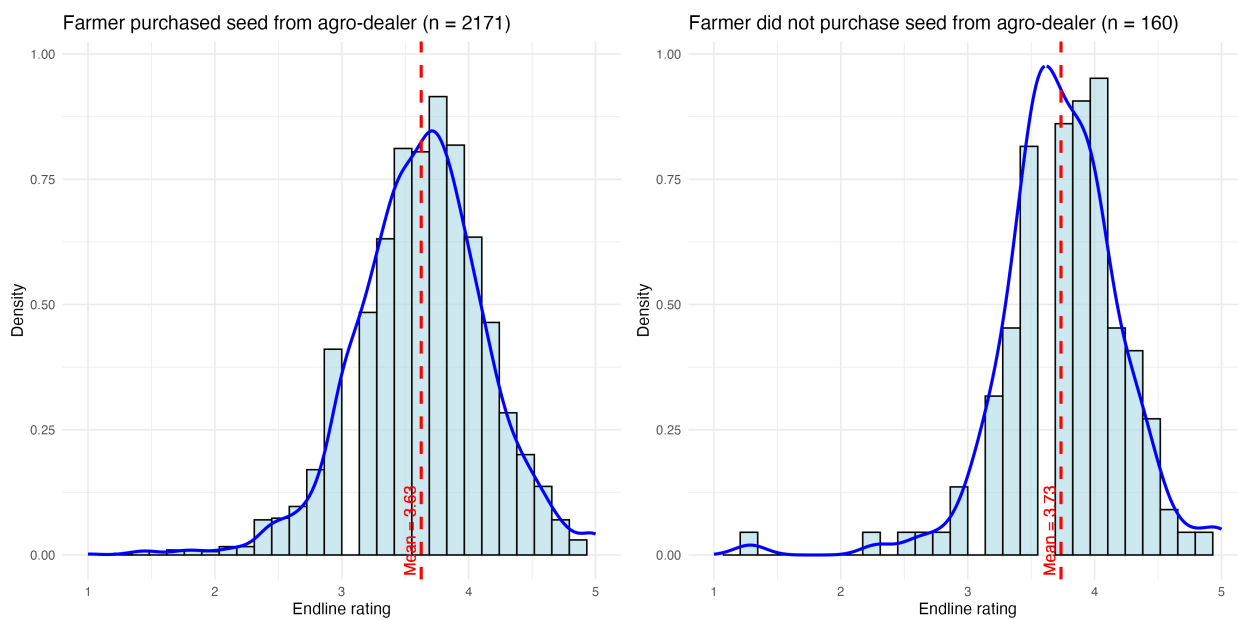


Figure A2: Comparison of ratings by purchase channel

Table A2: Correlating ratings and performance indicators - Agro-dealer level

|                                                                 | Ratings            |
|-----------------------------------------------------------------|--------------------|
| Agro-dealer only sells farm inputs                              | 0.205**<br>(0.078) |
| Index of labor-intensive seed handling practices <sup>1</sup>   | 0.164+<br>(0.091)  |
| Index of capital-intensive seed handling practices <sup>2</sup> | 0.156+<br>(0.081)  |
| Index of all seed handling practices <sup>3</sup>               | 0.205+<br>(0.116)  |
| Agro-dealer received seed related complaint from customer       | -0.108<br>(0.093)  |
| Shop received warning after inspection                          | -0.034<br>(0.086)  |

Note: <sup>1</sup>The index of labor-intensive seed handling practices contains 6 variables: whether seed is stored in dedicated area, in correct lighting, on correct surface, not in open containers, whether shop has no pest problem, cleanness and professionalism rating by enumerator.

<sup>2</sup>The index of capital-intensive seed handling practices contains 6 variables: whether roof is leak-proof, roof is insulated, walls are insulated, shop is ventilated, shop displays official certificate, expired seed is handled correctly.

<sup>3</sup>The index of all seed handling practices contains 12 variables: the ones included in the index of capital-intensive practices and the ones included in the index of labor-intensive practices.

Table A3: Correlating ratings and yield on randomly selected field - Farmer level

|                                                                | Yield in kg/acre <sub>endline</sub> |
|----------------------------------------------------------------|-------------------------------------|
| Index of farmer's seed ratings <sub>midline</sub> <sup>1</sup> | 144.971**<br>(8.564)                |
| Index of farmer's seed ratings <sub>endline</sub> <sup>1</sup> | 35.225**<br>(9.484)                 |

Note: <sup>1</sup>The index of farmer's seed ratings contains 6 ratings: general quality, yield, drought tolerance, pest/disease tolerance, speed of maturing, and germination.

## A.5 Treatment effects on agro-dealers' business activity

We examine treatment effects on agro-dealer business activity for two maize varieties: the most recently released hybrid, Longe 10H (Table A4) and the most recently released OPV, Longe 5 (Table A5). In addition to the general business outcomes described in Subsection 6.2, we include stock indicators. Agro-dealers reported how much seed of each variety was carried over from the previous season, how much was procured from suppliers, how much was lost or wasted, and how often stock-outs occurred.

Higher demand induced by the intervention should increase turnover, reducing carryover and increasing procurement of fresh seed. Higher turnover may also reduce losses and stock-outs.

At midline, we find no significant effects. By endline, however, all coefficients move in the expected direction, and the combined [Anderson \(2008\)](#) index shows a positive and statistically significant effect. This pattern is consistent with increased farmer engagement translating into higher agro-dealer business activity.

Table A4: Treatment effects on agro-dealers' business activity (Longe 10H)

|                                                      | <i>baseline</i>        |  | <i>midline</i>    |      | <i>endline</i>     |      |
|------------------------------------------------------|------------------------|--|-------------------|------|--------------------|------|
|                                                      | mean                   |  | treated           | obs. | treated            | obs. |
| Sales volume in kg <sup>§†</sup>                     | 288.384<br>(727.049)   |  | 0.236<br>(0.204)  | 256  | 0.352<br>(0.239)   | 242  |
| Sales price in UGX/kg <sup>§†</sup>                  | 6220.899<br>(1153.371) |  | -0.013<br>(0.026) | 194  | 0.010<br>(0.029)   | 187  |
| Revenue in mln UGX <sup>§†</sup>                     | 1.625<br>(3.839)       |  | 0.130<br>(0.123)  | 255  | 0.173<br>(0.136)   | 241  |
| Amount carried over in kg <sup>1§†</sup>             | 2.679<br>(12.137)      |  | 0.090<br>(0.215)  | 262  | -0.034<br>(0.134)  | 250  |
| Amount shop bought from provider in kg <sup>§†</sup> | 294.672<br>(741.810)   |  | 0.206<br>(0.213)  | 257  | -0.022<br>(0.253)  | 243  |
| Amount lost/wasted in kg <sup>§†</sup>               | 0.036<br>(0.405)       |  | 0.019<br>(0.097)  | 257  | -0.038<br>(0.041)  | 243  |
| Number of times per month shop ran out <sup>§†</sup> | 1.039<br>(1.575)       |  | -0.045<br>(0.133) | 192  | -0.205<br>(0.136)  | 185  |
| <b>Overall Anderson index</b>                        | 0.080<br>(0.437)       |  | 0.029<br>(0.070)  | 244  | 0.217**<br>(0.057) | 233  |
| Max. number of obs. <sup>2</sup>                     |                        |  |                   | 268  |                    | 254  |

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels; † indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

<sup>§</sup>Due to the skewness of this variable, the regression was run after an inverse hyperbolic sine transformation. Coefficient estimates can therefore be interpreted as percentage changes. The baseline mean column shows the untransformed variable.

<sup>1</sup>Amount carried over is low at baseline, as many agro-dealers report carrying no seed over from the previous season.

<sup>2</sup>The comparisons were only made for agro-dealers that had Longe 10H in stock at mid- or endline.

Table A5: Treatment effects on agro-dealers' business activity (Longe 5)

|                                                      | <i>baseline</i>       |  | <i>midline</i>    |      | <i>endline</i>    |      |
|------------------------------------------------------|-----------------------|--|-------------------|------|-------------------|------|
|                                                      | mean                  |  | treated           | obs. | treated           | obs. |
| Sales volume in kg <sup>§†</sup>                     | 389.492<br>(716.556)  |  | 0.304<br>(0.216)  | 261  | 0.316<br>(0.230)  | 259  |
| Sales price in UGX/kg <sup>§†</sup>                  | 3114.463<br>(409.328) |  | -0.015<br>(0.016) | 249  | 0.013<br>(0.022)  | 241  |
| Revenue in mln UGX <sup>§†</sup>                     | 1.193<br>(2.175)      |  | 0.111<br>(0.096)  | 261  | 0.114<br>(0.105)  | 258  |
| Amount carried over in kg <sup>1§†</sup>             | 4.312<br>(19.088)     |  | -0.092<br>(0.306) | 270  | -0.004<br>(0.155) | 263  |
| Amount shop bought from provider in kg <sup>§†</sup> | 431.451<br>(803.696)  |  | 0.253<br>(0.215)  | 262  | 0.289<br>(0.235)  | 260  |
| Amount lost/wasted in kg <sup>§†</sup>               | 1.756<br>(10.173)     |  | 0.031<br>(0.128)  | 266  | -0.055<br>(0.058) | 261  |
| Number of times per month shop ran out <sup>§†</sup> | 0.839<br>(1.509)      |  | 0.086<br>(0.101)  | 248  | -0.054<br>(0.126) | 237  |
| <b>Overall Anderson index</b>                        | 0.039<br>(0.401)      |  | 0.012<br>(0.062)  | 256  | 0.152*<br>(0.058) | 252  |
| Max. number of obs. <sup>2</sup>                     |                       |  |                   | 275  |                   | 269  |

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels; † indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

<sup>§</sup>Due to the skewness of this variable, the regression was run after an inverse hyperbolic sine transformation. Coefficient estimates can therefore be interpreted as percentage changes. The baseline mean column shows the untransformed variable.

<sup>1</sup>Amount carried over is low at baseline, as many agro-dealers report carrying no seed over from the previous season.

<sup>2</sup>The comparisons were only made for agro-dealers that had Longe 5 in stock at mid- or endline.

## A.6 Treatment effects on agro-dealers' seed performance

Moisture content – a pre-registered, objectively measured indicator of seed performance – shows a statistically insignificant negative treatment effect. Because not all agro-dealers had maize seed in stock, the realized sample size is smaller than planned: 232 observations at baseline, 179 at midline, and 265 at endline, compared with a target of 348.

To assess whether the null result reflects a true absence of effect or insufficient statistical power, we assume that the estimated treatment effect at endline – a 0.22 percentage point reduction in seed moisture content – represents the true effect. Figure A3 plots a power curve relating sample size to the probability of detecting this effect at the 5% significance level, given the observed baseline variability in moisture content. The curve shows that the null result is unlikely to be driven by the reduced sample size: even with substantially larger samples, an effect of this magnitude would remain undetectable.

We also examine additional indicators of seed performance, including binary variables for whether an agro-dealer received a seed-related customer complaint, a warning following an inspection, or had products confiscated during an inspection. These outcomes are available for larger samples than the indicators derived from purchased seed bags in Table 11. We find no treatment effects on these measures, providing further evidence that the intervention does not affect seed performance.



Figure A3: Power curve for seed moisture at endline

Table A6: Treatment effects on agro-dealers' seed performance (alternative proxies)

|                                                                 | <i>baseline</i>   | <i>midline</i>    |      | <i>endline</i>    |      |
|-----------------------------------------------------------------|-------------------|-------------------|------|-------------------|------|
|                                                                 | mean              | treated           | obs. | treated           | obs. |
| Shop received seed related complaint from customer <sup>†</sup> | 0.644<br>(0.480)  | -0.089<br>(0.057) | 306  | -0.039<br>(0.048) | 297  |
| Shop received warning after inspection <sup>†</sup>             | 0.317<br>(0.466)  | 0.005<br>(0.073)  | 291  | -0.009<br>(0.063) | 284  |
| Shop's products were confiscated after inspection <sup>†</sup>  | 0.145<br>(0.353)  | -0.027<br>(0.046) | 293  | -0.025<br>(0.036) | 285  |
| <b>Overall Anderson index</b>                                   | -0.007<br>(0.680) | 0.066<br>(0.100)  | 303  | 0.060<br>(0.087)  | 292  |
| Max. number of obs.                                             |                   |                   | 306  |                   | 297  |

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels; <sup>†</sup> indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

## A.7 Alternative index construction

We recompute all summary indices following [Kling, Liebman, and Katz \(2007\)](#) as a robustness check to the [Anderson \(2008\)](#) indices used in the main analysis. For farmers' perceptions of seed performance, the treatment effect becomes statistically insignificant when the index is constructed following [Kling, Liebman, and Katz \(2007\)](#). Results for farmers' adoption and sourcing decisions remain robust. Results for agro-dealers' business activity (general, Longe 10H, and Longe 5) are largely robust.

For farmers' harvest outcomes, the positive endline effect remains robust. At mid-line, however, the [Kling, Liebman, and Katz \(2007\)](#) index yields a significant negative coefficient. The harvest outcomes included in the index are mechanically related (e.g., yield, production, and revenue). The [Anderson \(2008\)](#) method down-weights highly correlated components, whereas [Kling, Liebman, and Katz \(2007\)](#) assigns equal weights and treats these outcomes as independent, which can amplify the influence of redundant information and generate a negative treatment effect. That is why we rely on the [Anderson \(2008\)](#) method, as pre-registered in our [pre-analysis plan](#), and use [Kling, Liebman, and Katz \(2007\)](#) only as a robustness check.

Results for agro-dealers' practices remain robust, as do all null results regarding agro-dealers' seed performance.

Table A7: Anderson (2008) vs. Kling, Liebman, and Katz (2007) indices

|                                                      | Anderson           |                               | Kling et al.       |                               |
|------------------------------------------------------|--------------------|-------------------------------|--------------------|-------------------------------|
|                                                      | <i>midline</i>     | <i>endline</i>                | <i>midline</i>     | <i>endline</i>                |
| Farmers' perceived seed performance                  |                    | 0.092 <sup>+</sup><br>(0.054) |                    | 0.088<br>(0.056)              |
| Farmers' adoption and sourcing decisions             | 0.087*<br>(0.042)  | 0.086*<br>(0.039)             | 0.096*<br>(0.041)  | 0.076*<br>(0.038)             |
| Agro-dealers' business activity (general)            | 0.197*<br>(0.092)  | 0.238**<br>(0.082)            | 0.189<br>(0.125)   | 0.247 <sup>+</sup><br>(0.132) |
| Agro-dealers' business activity (Longe 10H)          | 0.029<br>(0.070)   | 0.217**<br>(0.057)            | 0.060<br>(0.082)   | 0.232**<br>(0.076)            |
| Agro-dealers' business activity (Longe 5)            | 0.012<br>(0.062)   | 0.152*<br>(0.058)             | 0.105<br>(0.089)   | 0.160 <sup>+</sup><br>(0.081) |
| Farmers' harvest                                     | -0.061<br>(0.039)  | 0.097*<br>(0.041)             | -0.092*<br>(0.046) | 0.082 <sup>+</sup><br>(0.046) |
| Agro-dealers' practices                              | 0.375**<br>(0.113) | 0.178 <sup>+</sup><br>(0.106) | 0.272*<br>(0.118)  | 0.238*<br>(0.118)             |
| Agro-dealers' seed performance indicators            | 0.108<br>(0.103)   | 0.108<br>(0.090)              | 0.065<br>(0.111)   | 0.064<br>(0.087)              |
| Agro-dealers' seed performance (alternative proxies) | 0.066<br>(0.100)   | 0.060<br>(0.087)              | 0.062<br>(0.103)   | 0.059<br>(0.087)              |

Note: Columns (1) and (2) report differences between treatment and control groups, with standard errors below (clustered at the level of randomization), if the overall indices are constructed following Anderson (2008); columns (3) and (4) report differences between treatment and control groups, with standard errors below (clustered at the level of randomization), if the overall indices are constructed following Kling, Liebman, and Katz (2007); \*\*, \*, and + denote significance at the 1%, 5%, and 10% levels; larger indices indicate more desirable outcomes.